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A review on lifetime prediction of proton exchange membrane fuel cells system

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***Abstract*— The proton exchange membrane fuel cells (PEMFC) system is a promising eco-friendly power converter device in a wide range of applications, especially in the transportation area. Insufficient service life is one of the key issues that hinder its large-scale commercial application. Degradations of the materials, suboptimal operating procedures, and disturbances from the external environment and load all would have a terrible impact on the durability performance of PEMFC. Prognostic and health management (PHM) can estimate the remaining useful life (RUL) in advance and make the proper decisions at the right time to extend the service life of the system by condition-based maintenance. At the very least, the PHM can provide degradation information and reserve enough maintenance time for the operators to avoid some fatal damages. Analyzing the working principles and the degradation phenomena of the PEMFC system are the foundations of degradation prediction. Developing accurate and dynamic models to describe the degradation mechanisms, exploring accurate, efficient, and robust methods to realize the RUL prediction are some critical challenges of PHM nowadays, especially under dynamic operating conditions. This paper focuses on comparing different models for degradation mechanisms, discussing some key issues of PHM, and reviewing the degradation prediction methods.**

Keywords—Proton exchange membrane fuel cell, Degradation, Prognostic, Durability, Remaining useful life

NOMENCLATURE

Abbreviations

PEMFC	Proton exchange membrane fuel cells	FFNN	Feed-forward neural network
PHM	Prognostic and health management	ARNN	Autoregressive neural network
RUL	Remaining useful life	ELM	Extreme learning machine
PGM	Platinum group metals	GMDH	Group method of data handling
FCEV	Fuel cell electric vehicles	GA	Genetic algorithm
ICE	Internal combustion engines	GNNM	Grey neural network model
CBM	Condition-based maintenance	BPNN	Backpropagation neural network
MAF	Moving average filtering	RVM	Relevance vector machine
HIs	Health indicators	RNN	Recurrent neural network
OCV	Open-circuit voltage	KNN	K-nearest neighbor
EKF	Extended Kalman filter	ANOVA	Analysis of variance
RPLR	Relative power-loss rate	TDNN	Time delay neural network
LPV	Linear parameter varying	PAM	Physical aging model
		RPF	Regularized particle filter
		MPPT	Maximum power point tracking

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EoL	End of life	PSO	Particle swarm optimization
EoT	End of test	LSTM	Long short-term memory
EoP	End of prognostic	GRU	Gated recurrent unit
ECSA	Electrochemical catalytic surface area	ESN	Echo state network
LWR	Locally weighted regression	HIL	Hardware in the loop
μ-CHP	Micro-combined heat and power	IFORM	Inverse first-order reliability method
PH	Prediction horizon	NARX	Nonlinear autoregressive exogenous
ML	Machine learning	ANFIS	Adaptive neuro-fuzzy inference systems
SH	Satisfactory horizon	FDKF	Frequency domain Kalman filter
RMSE	Root mean square error	AUKF	Adaptive unscented Kalman filter
MAPE	Mean average percentage error	Symbols	
RA	Relative accuracy	p	Number of regressors
EIS	Electrochemical impedance spectroscopy	$\boldsymbol{t_s}$	Sampling time
ECM	Equivalent circuit model	$\boldsymbol{y_s}$	Measured stack voltage
SOH	State of health	$\boldsymbol{\hat{y}_s}$	Predicted stack voltage
UKF	Unscented Kalman filter	$\boldsymbol{t_{RUL}^{act}}$	Actual RUL
PF	Particle filter	$\boldsymbol{t_{RUL}^{pre}}$	Predicted RUL
ANN	Artificial neural network	m	Number of data points
GP	Gaussian process	H	Total prediction data points
SVM	Support vector machine	R_{over}	Overall resistance
DWT	Discrete wavelet transform	i_{L}	Limiting current

1. Introduction

The United Nations Department of Economic and Social Affairs (UN DESA) estimates that the world's population will rise to 9.77 billion in 2050 [1]. And the energy demand will increase by 50 % in the next 30 years according to the survey of Economics and Technology Research Institute (ETRI) which belongs to China National Petroleum Corporation (CNPC) [2]. Hydrogen (H_2) is considered to be the promising energy carrier of sustainable clean energy, and breakthroughs in hydrogen energy technology can relieve the pressure of energy and the environment [3]-[5]. Since the mid-20th century, fuel cells start gaining a lot of attention for they can realize the conversion from chemical energy (e.g., H_2) to electricity directly [6], [7]. Due to the twofold pressures of environmental issues (carbon emission) growth and fossil fuel (e.g., coal, oil) reserves being exhausted, the fuel cells are re-attracting more and more attention in the last two decades [8], [9]. Depending on the types of electrolyte, the fuel cells techniques can be divided into five kinds: proton exchange membrane fuel cells (PEMFC), alkaline fuel cells (AFC), phosphoric acid fuel cells (PAFC), molten carbonate fuel cells (MCFC), solid oxide fuel cells (SOFC) [10].

The PEMFC is the most popular and widely used one among the different types [11], [12]. The components of a single cell include bipolar plates, gas diffusion layers, catalyst layers, membrane, sealing gaskets, etc. For the single fuel cell, the protons in the hydrogen are released at the anode electrode (oxidation reaction) and transferred to the cathode electrode through the polymer membrane. In the same direction as the protons, the electrons are delivered from the anode electrode to the cathode electrode through the external circuit. Then, the reduction reaction is taken place on the cathode electrode by the electrons, oxygen, and protons. As a whole, the chemical energy in the hydrogen is directly transferred into the electrical energy, and the by-products are water and heat [13], [14]. To provide a usable

output voltage, a number of single cells are assembled and pressed together to make up a PEMFC stack [15]. The PEMFC system consists of the PEMFC stack and the corresponding ancillary equipment.

The PEMFC system has the advantages of being environmentally friendly, easy scaling between power and capacity, quick fueling (usually < 20 minutes), high peak efficiency (64 %), favorable power-to-weight ratio (0.86 kW/kg), quick start-up (< 30 s from -20 ℃), and low operating temperature (50-80 ℃) [16]-[18]. The PEMFC system is welcomed in the area of transportation (cars, buses, trucks, etc.), portable devices, backup equipment, and distributed generation [19]-[21]. Based on the statistical data of the E4Tech [22], the fuel cell industry shipped more than 1318.7 MW in 2020 in terms of total power, and the corresponding total unit shipments were more than 8.24×10^4 sets. The PEMFC has a steady development from 2016 to 2020 and dominates shipments by fuel cell type, both in capacity (1029.7 MW in 2020) and numbers (5.36×10^4 sets in 2020).

Even though the PEMFC system has the above-mentioned advantages, its large-scale commercialization is still hampered by the bottlenecks of high cost and low performance [23]-[25].

Cost: Based on the statistics of the United States Department of Energy (U.S. DOE), the current system cost (durability-adjusted) is 76 $/kW, and the ultimate target is 30 $/kW when the production is 100000 units/year [26]. After the evaluation of an automotive PEMFC system (80 kW), the noble metal catalysts and bipolar plates (e.g., graphite) are the largest cost components of the PEMFC stack [27]. The mass production and application of fuel cell components can help reduce the costs of membrane and gas diffusion layers to some extent. Reducing the content of Platinum group metals (PGM), seeking inexpensive catalyst (PGM-free) materials, and developing the materials of bipolar plates are some research focuses for decreasing the cost [28]. Besides, the hydrogen's production, delivery, and storage, the auxiliary equipment, and the drivetrain maintenance can introduce other important costs.

Performance: Based on the statistics of the U.S. DOE, the system's performance of current status versus (vs.) ultimate targets are: peak energy efficiency (64 % vs. 70 %), power density (640 W/L vs. 850 W/L), specific power (860 W/kg vs. 900 W/kg), durability in automotive drive cycle (4130 h vs. 8000 h), etc. [16]. Some directions of efforts to improve the performance of the PEMFC system are: exploring novel materials, developing hydrogen storage technologies, improving the catalyst activity, exploring the more excellent electrolytes, using the most advanced integration technologies, etc. [29]. Durability is the most important property of the PEMFC system especially for the transportation application [30], and it still has a large room to be improved when compared to the target value [31].

Degradation is one of the key reasons that lead to limited durability, and the degradation prediction based on some prognostic methods are helpful for avoiding some fatal damages [32], [33]. Nevertheless, the degradation prediction is a difficult task because of the multi-physics, multivariate, and multiphase characteristics of the PEMFC system. Generally, the power (or voltage) loss between the measured value and its initial value at a constant current is commonly used as the outward manifestation of degradation. The U.S. DOE has defined the 10 % voltage drop (current is 55% - 65% of rated value) metric for assessing the fuel cell durability of fuel cell electric vehicles (FCEV). The durability performance of PEMFC should be equivalently durable and reliable to compete with internal combustion engines (ICE), and this is the reason why the U.S. DOE regards 8000 h as the durability target for FCEV.

The PEMFC system is a complex intersection of many research fields such as electrochemistry, thermodynamics, mechanical, hydromechanics, etc. [34]-[36]. The degradation phenomena of the PEMFC system happen on different timescales (from μs to h) and space scales (from nm to m). The material aging, structure designing and fabrication process of different components, external disturbances, and operating conditions would affect the durability

performance of PEMFC. Once a PEMFC stack is assembled and in use, the degradation will occur during its whole life until the breakdown [37], [38]. Exploring the degradation characteristics helps the users to analyze the output performance and improve the service life of a PEMFC system. The degradation phenomena can be broadly classified into mechanical degradation and chemical degradation [39], [40].

Mechanical degradations are mainly caused by the non-uniform mechanical stresses which may result from material's congenital defects, or the structure designing and fabrication process of different components. The physical dimensions of the components are usually shrunken at low temperature (or low relative humidity) or swelled at high temperature (or high relative humidity). The pressure difference between the anode and cathode chambers also contributes to mechanical degradation. The uneven stresses are also present at the contact surface between two components which may produce small perforations and tears. Increasing the number of cells can improve the output voltage of the PEMFC stack. Nevertheless, this multiple cell stacked structure aggravates the inhomogeneous distribution of mechanical stresses.

Chemical degradations are mainly caused by the material's natural aging and external disturbances. In detail, the corrosion of bipolar plates, oxidation of the carbon-based material, dissolution of catalyst, and polymer decomposition are the common material aging factors. The temperature, relative humidity, hydrogen ion concentration (pH) value, and poisonous gas (e.g., sulfur oxides and carbon monoxide (CO)) are the common external disturbances [41], [42]. The temperature, relative humidity, and reactants' concentration are heterogeneous in a single cell's surface area, and it would be more serious when the cell has a large surface area [43]. This nonuniformity may lead to a high current density (high degradation speed) in some areas and a low current density (low degradation speed) in other areas.

Generally speaking, the longer the PEMFC stack is operated especially in transient or varying load profiles operating conditions, the more serious is the degradation and the shorter is the lifetime [44]. Thus, the service life of the PEMFC system for automobiles is even more challenging because they have to face the operating conditions of start-stop cycles, load cycling, empty and heavy load current, etc. Sufficient evidence shows that these complex operating conditions increase the degradation rates of PEMFC especially the load cycling [45]. In summary, the degradation speed is different from cell to cell, and the durability characteristics of different components are also varying in the same cell. It is a great challenge to fully identify all deterioration mechanisms or to completely understand the degradation phenomena of the PEMFC system.

2. Prognostic and health management of PEMFC system

Prognostic and health management (PHM) has the ability to estimate the degradation state based on the past working profiles and the current operations, and then to minimize repair cost and extend the lifespan based on the condition-based maintenance (CBM) [46], [47]. Prognostic is the key process of PHM, and the International Organization for Standardization (ISO) has defined the concept of prognostic: prognostic is an estimation of time to failure and the probability of one or more existing and future failure modes [48]. In the PEMFC system, be aware, the prognostic does not address catastrophic stack failure. The main processes of the prognostic in four steps are shown in Fig. 1 [49].

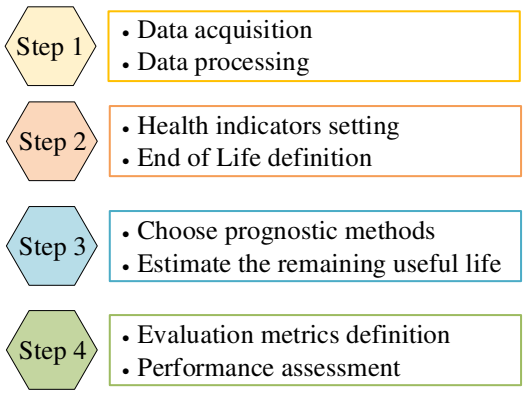


Fig. 1. The stages of the prognostic process.

2.1 Data acquisition and processing

The durability data of the PEMFC system are scarce especially at the variable load current and the long-term period [50]. This is because the testing process is rather complicated, and some unscheduled conditions also have serious effects on the testing process such as the sudden startup and shutdown, transient operation, availability of the facility, etc. It is even more difficult for general researchers to acquire the durability data due to the lack of a safe and reliable testing environment. To promote the development of PHM, the "IEEE PHM 2014 Data Challenge" ("Data Challenge" for short) was launched by the IEEE Reliability Society, FCLAB research federation, FEMTO-ST Institute, and the Laboratory of excellence ACTION [51]. Two new PEMFC stacks were tested and recorded for more than 1000 h: the first stack was operated under the steady-state (SS) condition (FC1, 1050 h), and the second stack was operated under the quasi-dynamic (QD) condition (FC2, 1020 h). The constant load current of 70 A is imposed on the aging test of FC1, and a triangular ripple current of 7 A at different frequencies (e.g., 5 kHz) is superimposed to the constant current of 70 A in the aging test of FC2. Most of the subsequent research works are based on the open data of Data Challenge. Nevertheless, these two open datasets are tested in the laboratory and under ideal circumstances. They are defective to simulate the real operating conditions in practice because the mission profiles such as the transportation applications are time-varying which are determined by the driving behavior, road condition, energy demand, etc. Testing the PEMFC system under dynamic operating conditions or even testing the on-road vehicles which are equipped by the PEMFC system have more practical meanings.

The data acquisition is better be implemented at the stack level from a practical point of view. In this process, the electrical signals (voltage, current, power, etc.) and physical signals (temperature, pressure, mass flow, etc.) are measured by the installed sensors. Inevitably, the disturbances (e.g., caused by data acquisition devices and profile vibration) and fault signals would be monitored by the sensors along with the degradation-related features in practices. These nondegenerate signals will have a strong impact on the results if used directly, and they should be pre-processed before using. Filtering is one of the frequently used data processing technologies in the literature, e.g., the loess filter [52], [53], Gaussian-weighted moving average filter [54], the moving average filtering (MAF) method [55], [56], and Gaussian Kernel-based smoother method [57].

The filtering coefficients have a significant effect on the results for each filtering method, and they are difficult to be decided in practice. For example, a too-large moving window of MAF may distort the original signal, and the periodic or global degradation information may be lost. During the degradation-related features extraction process, a

too-small moving window of MAF may maintain lots of noise. The trial-and-error is still the common coefficients setting method nowadays. In addition to the filtering technology, other methods such as locally weighted regression (LWR) can also be used for the data pre-processing, and the results of different smoothing scales are compared in [58].

The traditional data pre-processing technologies cannot identify the degradation features that belong to which component of the PEMFC stack. Developing advanced methods to separate the degradation-related features into different timescales and space scales that correspond to the degradation of different components with their corresponding physical background is a promising research focus. Besides, combining the diagnostic and prognostic together has the potential to classify the degradation features, fault information, and disturbance signals separately.

2.2 Health indicators and end of life

2.2.1. Health indicators

Health indicators (HIs) can be used to evaluate the degradation state of the PEMFC system, and the efficient HIs are the foundation of lifetime prediction [59]. The fuel cell "voltage" and "power" are two commonly used static HIs at the cell/stack level, and they are the external manifestation of the degradation at the component level [60]. The output voltage and power are easy to be measured. Besides, they are usually used in the control loops. At a constant load current, the output voltage and power have a nonreversible monotonic decreasing tendency as a whole regardless of the causes of degradation (some recovering episodes may occur) [61].

In addition to the single HI ("voltage" or "power"), some hybrid HIs are also introduced under static operating conditions [62], [63]. The HI of "geodesic distance" is synthesized by the voltage, power, and overall resistance in [62]. Another hybrid HI has combined the stack voltage, overall resistance, and open-circuit voltage (OCV) to improve the prediction accuracy [63]. Nevertheless, multiple indicators would increase the computational complexity and the weights of different indicators are difficult to be decided. In practice, the static HIs are synthetically affected by the mission profiles and deterioration factors. Their abilities to indicate the degradation state would be limited under dynamic operating conditions. Since it is difficult to identify the effects whether caused by degradation factors or by the load current.

To indicate the degradation state under the dynamic operating conditions, the dynamic HIs such as "electrochemical catalytic surface area (ECSA)" [64], "extended Kalman filter (EKF)-based degradation factor (DF)" [65], "virtual steady-state voltage (VSV)" [66], and "relative power-loss rate (RPLR)" [67] have been put forward by some scholars.

The relationship between ECSA and cell voltage is built, and the cyclic load profile is used to validate its effectiveness [64]. The cyclic voltammetry allows to measure the ECSA but cannot be implemented online as it is a laboratory characterization, and this is the main constraint for its practical use. For "EKF-based DF" in [65], two monotonous variables (overall resistance R_{over}, and limiting current i_L) are calculated by the periodically measured polarization curves, and then they are rewritten by the linearization equations. Nevertheless, the linearization hypothesis of the parameters is imprecise for the nonlinear PEMFC system. For "VSV" in [66], the voltage fragments are represented by group models of linear parameter varying (LPV), and the HI is identified from the LPV models. The extraction process of "VSV" is computational complexity, and this dynamic HI is more suitable for regular mission profiles. For RPLR in [67], it is extracted from the polarization curves and the measured electrical signal

(voltage and current). The RPLR can weaken the effects of variable load profiles and has a monotonically decreasing characteristic. Compared with "EKF-based DF", the RPLR is not restricted by the large time interval of R_{over} and i_L (about one week) and the linearized approximation. The computing burden of RPLR is lighter than the "VSV" because the calculation of PRLR is only based on the arithmetic operation. Nevertheless, the identification of "VSV" is based on the high-dimensional nonlinear radial basis function (RBF). Exploring the simple, online, and applicable to all operating states HIs is a challenging task at present [68]. Classification of existing typical HIs is summarized in Tab. 1.

Tab. 1 The classification of typical HIs.

Operating state	HIs	Advantages	Disadvantages
Static	Voltage, Power	Easy to be measured online; Used for the control loop.	Disturbed by the load; Cannot be used for dynamic states.
	Hybrid	Accurate; Including physical degradation factor.	High computational complexity; Difficult to decide the weights.
Dynamic	ECSA	Building models between ECSA and voltage; Cyclic voltammetry.	Offline; Interrupt the working of the fuel cell.
	EKF-based DF	Simple (linearization hypothesis); Based on polarization curves.	Imprecise; large time interval (about one week)
	VSV	A group of LPV models; accurate; online.	High computational complexity; More suitable for load cycling.
	RPLR	Simple; Online, Suitable for different operating states.	Current effects can be weakened but not totally eliminated.

2.2.2. End of life definition

The definition of the real end of life (EoL) for the PEMFC system is still an open question. As stated in the introduction, the metric of 10 % voltage drop for the vehicle application was defined by U.S. DOE to assess fuel cells' durability from different original equipment manufacturers [69]. The metrics are also different in multiple applications such as 20 % for micro-scale (1-10 kW) combined heat and power (CHP), 10 % for medium-scale (100 kW-3 MW) CHP, and 20 % for portable devices.

Thus, reaching an absolute consensus on the EoL threshold to measure the degradation state for different applications is impossible. And it is usually defined by the users themselves. On the one hand, the experimental tests in most of the papers are different from the testing conditions of the U.S. DOE. On the other hand, the degradation prediction methods can also make a fair comparison in the same EoL threshold no matter what its value is. For example, the EoL thresholds are {6, 10, 15} % of initial power in [53], the {3.5, 4.0, 4.5, 5.0, 5.5} % loss of initial power have been used as EoL thresholds in the Data Challenge [51]. In the Data Challenge, the EoL thresholds and the failure thresholds are the same to make it easier to calculate the remaining useful life (RUL) easily and validate the prediction methods simply. The EoL-related nomenclatures are defined as [70]:

Failure time means the PEMFC is broken and doesn't produce power anymore. When the PEMFC stack reaches the failure threshold, it must be fixed or replaced by a new one.

EoL time (t_{EoL}) means the produced power of the PEMFC stack cannot fit anymore the application requirement, and it still has the ability to work. Considering the economic factors, the stack should be replaced when it reaches the

EoL threshold in most conditions. The t_{aEoL} and t_{pEoL} are the time when the actual signal and the prediction signal firstly reaches the EoL threshold separately.

End of test (EoT) time (t_{EoT}) means the time when the durability test is finished.

End of prognostic (EoP) time (t_{EoP}) means the time when the prognostic method is finished.

Prediction horizon (PH) is defined as the time interval between the time when prediction begins (t_{pre}) and the t_{EoP}. If the prediction stopped at the EoL threshold, the PH is equal to the time interval between the t_{pre} and the t_{pEoL}.

The EoL can be regarded as a special condition of EoT [67]. The definition of EoL and EoL-related nomenclatures are shown in Fig. 2. PH1 is the prediction horizon of curve ①, and PH2 is the prediction horizon of curve ②. In general, a longer PH is more useful for the users.

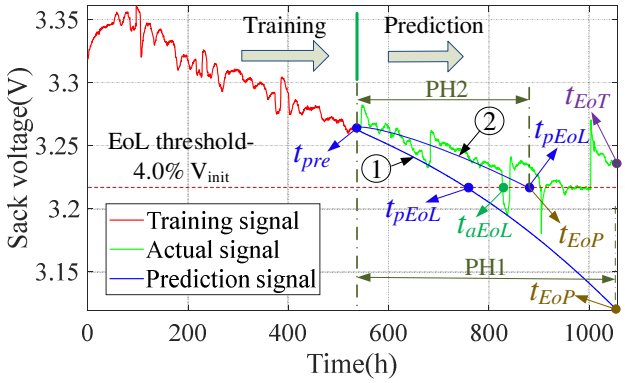


Fig. 2. The definition of EoL and EoL-related terminologies [56].

2.3 Prognostic methods and remaining useful life

2.3.1. Prognostic methods

Various prognostic methods have been proposed to estimate the degradation state of the PEMFC system [71], [72]. According to whether mathematical/analytical models exist, these strategies can be generally categorized into **model-based** and **data-driven** types. Through the analysis of the literature, a classification of prognostic methods is made in Fig. 3.

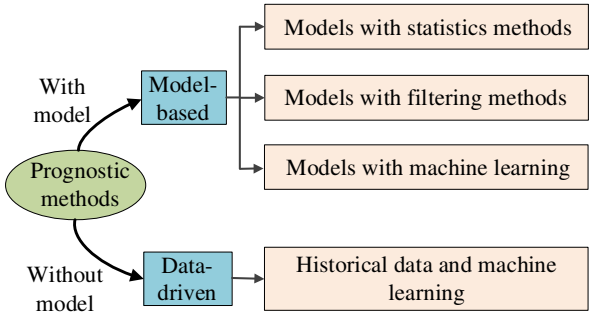


Fig. 3. The classification of prognostic methods.

Model-based type is the most general and intuitive approach because it can reveal some physical mechanisms and numerous models have already been explored [36], [41]-[45], [73]-[82]. The principle of the model-based methods can be described as: building the models (e.g., empirical, semi-empirical, and physical) of the PEMFC system. And then the degradation prediction is realized by the techniques of statistics [83], [84], filtering [57], [61],

[64], [65], [86]-[88], [90]-[112], machine learning (ML) methods [113]-[127], etc.

Unlike the model-based type, the data-driven approach can be regarded as one "black box" that is used to construct high-order nonlinear relations between the input and output signals [52]-[56], [58], [67], [128]-[163]. The future degradation state is based on the historical degradation tendency. The precise mathematical/analytical models are no longer required. In this condition, the degradation mechanisms of the components, cells, or stacks do not need to be fully revealed. The degradation phenomena caused by different components can be considered as a whole. Data-driven methods are usually used in cases where the degradation process is difficult to be modeled or the data are easy to be acquired.

Some scholars called the approaches based on both models and filtering/ML the hybrid methods which try to combine the advantages of model-based and data-driven together [94], [114], [116], [120], [124]. In most cases, the accuracy of models always limits the overall performance of the hybrid methods. In this paper, the hybrid method is also classified to the model-based category since the models are necessary in this case.

Exploring more efficient prognostic methods is one of the research focuses in the next step, and the performance such as accuracy and efficiency of the methods are also needed to be further improved. Most of the literature has realized the prognostic offline. Nevertheless, online prediction is more interesting in practice. In the long run, more and more methods that can realize the lifespan's online prediction should be proposed and tested.

2.3.2. Single-step and multi-step ahead prediction

It is necessary to set the prediction step size before implementing all prognostic methods because the prediction step is one of the factors affecting the method's prediction accuracy. Based on the size of the prediction step, the prognostic methods in the degradation prediction of the PEMFC system can be classified into the single-step (or one-step) ahead and multi-step ahead types. Taking the time sequence of stack voltage as an example, the sampling points can be expressed as $\{(t_1, y_1),\cdots,(t_s, y_s)\}$. Where t_s represents the sampling time, y_s represents the stack voltage at the time t_s [157].

Single-step ahead: Based on the historical set of regressors $\{(t_{s-p+1}, y_{s-p+1}),\cdots,(t_s, y_s)\}$, the single-step ahead prediction is implemented to estimate the voltage value of the next step $(t_{s+1}, \hat{y}_{s+1})$. Where p is the number of regressors used, i.e., the number of past discrete values used for prediction. For example, the set of regressor is $\{(t_{s-2}, y_{s-2}), (t_{s-1}, y_{s-1}), (t_s, y_s)\}$ when p = 3. When the system reaches time t_{s+1}, the voltage y_{s+1} would be measured and added to the historical time sequence $\{(t_1, y_1),\cdots,(t_{s+1}, y_{s+1})\}$. Based on the data of $\{(t_{s-p+2}, y_{s-p+2}),\cdots,(t_{s+1}, y_{s+1})\}$, the single-step ahead prediction is implemented again to estimate the voltage value of the next step $(t_{s+2}, \hat{y}_{s+2})$.

Multi-step ahead: The multi-step ahead prediction can be realized by different approaches, and the iterative approach which is based on the single-step ahead prediction process is the simplest one [164]. At time t_s, the data of $\{(t_{s-p+1}, y_{s-p+1}),\cdots,(t_s, y_s)\}$ are used to estimate the voltage value of the next step $(t_{s+1}, \hat{y}_{s+1})$. At time t_{s+1}, the predicted voltage $\hat{y}_{s+1}$ would be added to the historical time sequence $\{(t_1, y_1),\cdots,(t_{s+1}, \hat{y}_{s+1})\}$. The data of $\{(t_{s-p+2}, y_{s-p+2}),\cdots,(t_{s+1}, \hat{y}_{s+1})\}$ are used to estimate the value of the next step $(t_{s+2}, \hat{y}_{s+2})$. By that analogy, the future voltage of $\hat{y}_{s+3}$, $\hat{y}_{s+4}$, …, $\hat{y}_{s+H}$ can be estimated, and index H denotes the total number of prediction data points. Nevertheless, the iterative approach suffers from propagation error, i.e., the prediction error increases with H.

In addition to the prediction step, the regressor number p and algorithm types would also affect the prediction

accuracy. Taking $\{f, \vartheta\}$ represents the trained models (or "black box") of single-step ahead prediction, $[f]$ represents the algorithm, and $[\vartheta]$ represents its parameters. The prediction structures of single-step ahead and multi-step ahead are shown in Fig. 4.

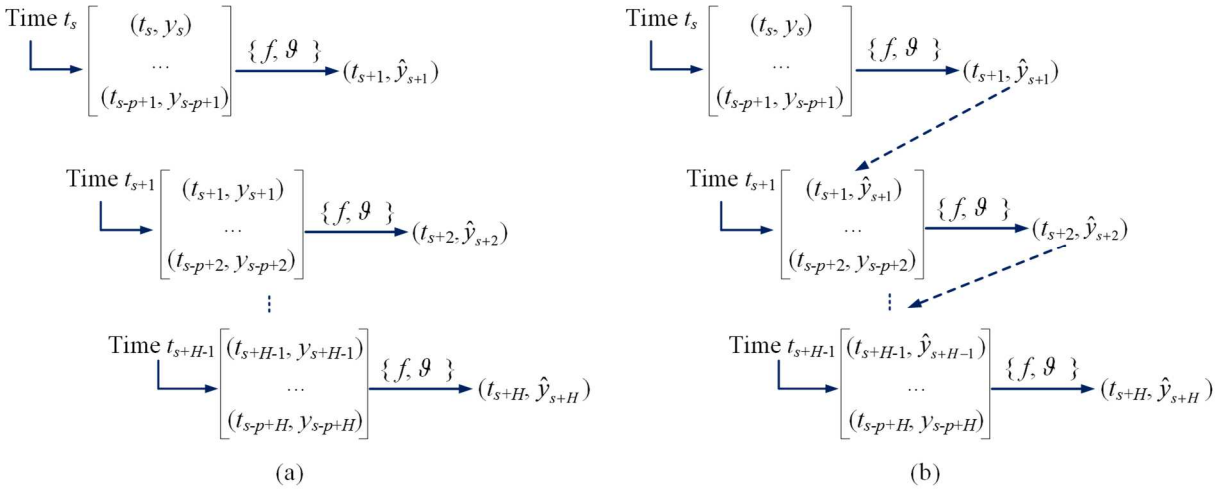


Fig. 4. Different prediction structures: (a) single-step ahead prediction, and (b) multi-step ahead prediction.

The PEMFC system typically has a lifetime of several thousand hours. Most of the time, the users want to know the degradation tendency in the following hundreds of hours. The time interval of single-step ahead prediction is usually in tens of minutes or several hours, and this cannot provide enough time for the users to take some maintenance actions in advance. Hundreds of hours, or at least dozens of hours predictions of degradation state in advance are more meaningful for the users. Thus, the multi-step ahead prediction is more helpful for the operators if their goal is to extend the lifespan.

In the multi-step ahead prediction structure, the time between t_s and t_{s+H} is regarded as the PH in this paper. There do not need the new measured data during the PH. Then the short-term, mid-term, and long-term predictions are classified by the length of PH. The short-term indicates that the prediction process is finished in several steps or dozens of steps and the PH is typically from a few minutes to several hours. The mid-term indicates that the prediction could last for dozens or hundreds of steps and the PH is typically from dozens of hours to over a hundred hours. The long-term prediction signifies that the overall RUL could be evaluated and the PH is in hundreds or even thousands of hours. In practical applications, the mid-term and long-term are more useful and preferred since they can provide more maintenance time for the users to extend the service life of the PEMFC system. The classification of prediction types by PH is shown in Tab. 2.

Tab. 2 The classification of prediction types by PH.

Types	PH
Short-term	PH ≤ 24 h (one day)
Mid-term	24 h (one day) < PH ≤ 168 h (one week)
Long-term	PH > 168 h (one week)

2.3.3. Remaining useful life

The main objective of prognostic is to predict the RUL which is commonly regarded as the time before a certain amount of power loss is reached [51]. More specifically, the actual (or target) RUL (t_{RUL}^{act}) can be defined as the time interval between t_{pre} and t_{aEoL}. The predicted RUL (t_{RUL}^{pre}) can be defined as the time interval between t_{pre} and the time when the prediction signal firstly reaches the t_{pEoL}. In general, there are two cases for the long-term prediction results: underestimation and overestimation. **Underestimation** means the predicted RUL is smaller than the actual RUL, and this is an early prediction (① in Fig. 2). **Overestimation** means the predicted RUL is greater than the actual RUL, and this is a late prediction (② in Fig. 2).

In practice, good performance of estimation means the underestimation of RUL (early warning). Underestimation is more meaningful and jeopardizes less to the operation of the system because it could warn the users to adjust the operating conditions or replace the PEMFC in advance and then prevent a fatal incident. Nevertheless, the overestimation does not have the warning functions because the estimated RUL would exceed the actual value. The RUL versus (vs.) time plot can be obtained by testing the RUL at different training lengths [70]. An example of RUL vs. time plot is shown in Fig. 5 (the actual RUL is the blue solid line).

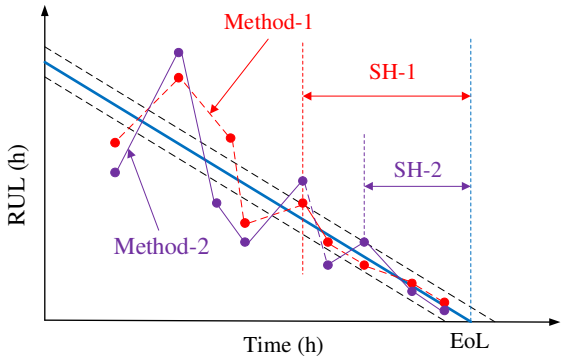


Fig. 5. The RUL vs. time plot with 5 % error bounds [67].

The satisfactory horizon (SH) is defined as the duration time when the prediction is located in an allowable error bound (e.g., 5 %) around the true RUL to the EoL (black dotted lines in Fig.5). The SH is more efficient than PH to evaluate the method's prediction ability, and a longer SH means that the prediction accuracy meets the requirement at an early stage. In Fig. 5, the SH-1 of method-1 is bigger than SH-2 of method-2, and it means that method-1 is more accurate than method-2 in the RUL prediction of the PEMFC system. The SH is regarded as PH in [67].

2.4 Evaluation metrics and performance assessment

Some effective evaluation metrics should be defined to quantitatively assess the prediction performance. The root mean square error (RMSE) and mean average percentage error (MAPE) are two commonly used criteria in the literature, and they can be used to estimate the sum error between the actual value and the predicted value quantitatively.

$$\begin{cases} \mathrm{RMSE} = \sqrt{\dfrac{1}{m}\sum_{i=1}^{m}(y_i^{pre} - y_i^{act})^2} \\ \mathrm{MAPE} = \dfrac{1}{m}\sum_{i=1}^{m}\left|\dfrac{y_i^{pre} - y_i^{act}}{y_i^{act}}\right| \end{cases} \tag{1}$$

Where y_i^{act} is the actual value, y_i^{pre} is the predicted value, and m is the number of data points.

The criterion of percent error on RUL estimation ($\%Er_{FT}$) was defined in the Data Challenge [51], and it is used to evaluate the error between the actual RUL (t_{RUL}^{act}) and the predicted one (t_{RUL}^{pre}).

$$\%Er_{FT} = \frac{t_{RUL}^{act} - t_{RUL}^{pre}}{t_{RUL}^{act}} \times 100 \tag{2}$$

The score of the estimation accuracy (A_{FT}) at each EoL threshold is defined as

$$A_{FT} = \begin{cases} exp^{-\ln(0.5)\cdot(\%Er_{FT}/5)} & if \ \ \%Er_{FT} \le 0 \\ exp^{+\ln(0.5)\cdot(\%Er_{FT}/20)} & if \ \ \%Er_{FT} > 0 \end{cases} \tag{3}$$

During the calculation, the underestimation (i.e., case of $\%Er_{FT} > 0$) has a slight deduction, and the overestimation (i.e., case of $\%Er_{FT} \le 0$) has a severe deduction. The final score of the whole predictions at different EoL thresholds is defined as the mean of all A_{FT}

$$\mathrm{Score}_{RUL} = \frac{1}{5}\sum\nolimits_{FT}(A_{FT}) \tag{4}$$

Winners of the Data Challenge are the participants (from universities and industries) who get the highest scores. In addition to the above-mentioned criteria, some other criteria have been used such as the "$\alpha - \lambda$ accuracy" [65], relative accuracy (RA) [127], coefficient of determination (R^2) [142], etc.

Based on the description above, the PH and SH can be regarded as the performance evaluation criteria. Usually, the longer the PH and SH, the better the prediction ability. Besides, the robustness, convergence rate, computational complexity, time, and memory of the prognostic methods are also the commonly used performance metrics. Based on the hierarchical design concept of the prognostic metrics [70], these criteria can be classified into three levels:

Level 1: Make a rough estimate based on PH, SH, $\alpha - \lambda$ accuracy, etc.

Level 2: Make a precise estimate based on RMSE, MAPE, R^2, RA, $\%Er_{FT}$, etc.

Level 3: Other auxiliary estimates based on robustness, convergence, computational complexity (e.g., time and memory), etc.

Combining these criteria is a more proper choice for the users to evaluate the prediction performance. Meanwhile, exploring other criteria to evaluate the prediction performance comprehensively in the long term is one of the research focuses in the next step.

3. Comparative study on degradation prediction of PEMFC system

3.1 Model-based methods

The degradation state of the PEMFC system can be modeled by physical, empirical, or semi-empirical equations, and the model's accuracy would have a direct impact on the prediction results. To improve the modeling accuracy,

signal processing, statistical techniques, filtering, and ML methods have been used to assist the modeling process.

3.1.1. Models for degradation prediction

Building a degradation model is fundamental to the model-based method. The polarization curve ($i - U$) can be used to reflect the static performance of the PEMFC. Besides, the electrochemical impedance spectroscopy (EIS) can also be used to analyze the dynamic characteristics of the PEMFC for the impendence is tested under different frequencies. After building the models to fit the polarization curves and EIS curves, the degradation-related parameters can be extracted from the models [73]-[82]. Then, the degradation state can be evaluated by predicting the tendency of these parameters [83], [84].

A pattern recognition method is proposed by Onanena *et al*. to predict the lifetime of the PEMFC system [73]. An empirical model is used to extract the static features from polarization curves, and both expert knowledge and parametric models are used to extract the dynamic features from EIS. The parameters which are more correlated to the degradation are selected to estimate the lifetime.

Based on the EIS measurements and linear regression technique, the impedance behavior estimation is realized in [74] and [75]. With the same models in [73], the mathematical equations are used to model the real part of EIS, and a high-order polynomial regression (5-order) is used to model the imaginary part of EIS in [74] by Vianna *et al*. Besides, the gradient descent approach is utilized to optimize the parameters of the mathematical equations, and the leave-one-out method is adopted to decide the order of polynomial regression. Based on the same datasets of [74], the equivalent circuit model (ECM) and voltage-resistance models are combined to realize the impedance prediction in [75] by Taejin *et al*. The ECM with 10 parameters is explored to express the physical characteristics of the PEMFC system, and then 4 parameters which can reflect the degradation phenomena are selected by the model reduction method. These models in [74] and [75] are tested by 2 datasets of Data Challenge [51], and they are the two winners of state of health (SOH) prediction from the industrial and academic participants. Their final scores are 7.8932×10^{-6} and 5.3177×10^{-6} separately.

Similar to [73], a prognostic-oriented model is proposed by Lechartier *et al.* [76]. The physical-based model is composed of two parts based on the measurements of polarization curves and EIS. The static part is built by the physical mechanism (Butler-Volmer law) of the PEMFC system, and the dynamic part is built by ECM. Finally, the proposed model is tested on the datasets of Data Challenge.

Two numerical models are proposed by Robin *et al*. to describe the Pt surface degradation under the nominal and power cycle operating conditions [77]. A physical model which focuses on the ECSA is proposed by Polverino *et al*. [78]. The activation losses in this model are represented by Tafel's equation, and the exchange current density is a function of ECSA. After fully understanding the decreasing rate of ECSA, the degradation model about the cell voltage is formulated. Finally, the degradation state can be estimated by comparing the current ECSA with its initial value. Besides, the effects of current density (0.05 A/cm^2-0.5 A/cm^2) and temperature (30 ℃-80 ℃) on the voltage decay are also analyzed. Results show that the voltage decay has a high speed when the current density is at the lower (0.05 A/cm^2) and higher levels (0.5 A/cm^2). And the voltage decay speed is decreasing with the increase of temperature (maximum 80 ℃).

Based on the vehicular operation records and the test results in the laboratory, Chen *et al*. built a model to realize the online lifespan prediction which considers the effects of the environmental and voltage degradation [79]. In

practice, the economic lifespan can be regarded as the service time when the average cost of the PEMFC system reaches the minimum value. And the total consumption in the proposed model is made up of manufacturing cost, hydrogen consumption, and attachment consumption (air compressor, cooling water, electronic control equipment, etc.). At last, the average cost of the PEMFC system and the economic lifespan are analyzed by considering different hydrogen prices and stack manufacture costs.

After a deep literature review on the degradation at different levels (component, cell, and stack), a semi-empirical aging model is proposed by Jouin *et al*. [80]. This model can describe the power behavior accuracy in its whole lifetime, and 4 datasets under the SS, QD, and dynamic operating conditions are used to validate the proposed model. The coefficients of correlation are higher than 0.96 in these 4 datasets. Similar to [80], an empirical voltage degradation model is proposed by Zhang *et al*. [81]. And both the aging terms and the mission profiles are used to build the voltage model.

A voltage model for lifetime prediction is proposed by Hu *et al*. [44]. The temperature fluctuation and sensor errors are considered in building the degradation model under the dynamic operating condition for a city bus. The degradation of ECSA is modeled by the form of an exponential function. The 14-day training data show that the deviation of the predicted voltage is less than 1 %.

Recently, a semi-empirical model is built based on the polarization curves by Ou *et al*. [82]. Different from the linearization hypothesis of overall resistance R_{over} in [44] and[65], the R_{over} is fitted by the linear plus single-term exponential function. The degradation form of ECSA is the same with [44], and a constant recovery factor is used to describe the recovery process after the shutdown. Two dynamic datasets with about 500 hours' durability are used to verify the improved model, and this model has a higher accuracy based on the 5-step ahead prediction results.

3.1.2. Models with statistics techniques

The model-based method of regime switching vector autoregressive (RSVAR) is proposed by Hochstein *et al*. to realize the RUL prediction [83]. The changing trend of the stack voltage in the next step is represented by the form of probability based on the first-order Markov model. The method is validated by the 2 datasets of Data Challenge in the long term (PH = 500 h) and gets the highest score (0.3592) from the industrial participants.

The error correction grey model (ECGM) is introduced in [84] by Zhou *et al*. A correction factor is considered to eliminate the inherent deviation of the typical grey model (GM). Besides, a trigonometric residual modification is used to describe the uncertainty of cyclic fluctuation. Based on the 2 datasets of Data Challenge (PH $\leq$ 495 h), the ECGM has better performance than GM in prediction accuracy, calculation efficiency, and robustness.

3.1.3. Models with filtering methods

The extended Kalman filter (EKF), unscented Kalman filter (UKF), and frequency domain Kalman filter (FDKF) are some improved structures of the Kalman filter (KF). The core assumption of KF is the propagation of a Gaussian random variable through the system dynamics. In EKF, the nonlinear state distribution is approximated by the first-order linearization. Nevertheless, the EKF has the disadvantages of parameter sensitivity and computation complexity. UKF can address the nonlinear problems by using a deterministic sampling method, i.e., a set of so-called sigma-points are sampled from the expectation and the covariance of the random vector [85]. To improve the approximation accuracy, the high-order nonlinear expansion of the Taylor series is used to replace the first-order linearization. Compared with EKF, the UKF needn't calculate the derivative of the Jacobian matrix, and it has a light

computational complexity. The FDKF achieves a good tradeoff between convergence rate the accuracy. The particle filter (PF) is based on the Monte Carlo principle. It can avoid the linearity and Gaussian noise assumption of KF, and predict the posterior probability density function (PDF) of the degradation state. The PF has received more and more attention for its excellent abilities to deal with uncertainties, arbitrary distributions, and nonlinearities. These filtering approaches (EKF, UKF, FDKF, and PF) and their variants have been applied to the degradation prediction of the PEMFC system. The prognostic methods by models and filtering methods are shown in Fig. 6.

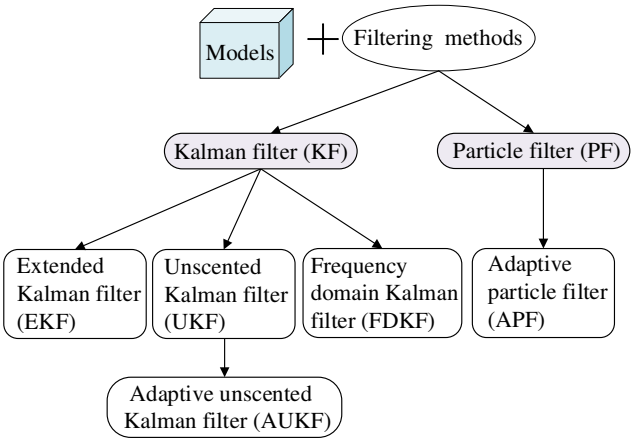


Fig. 6. The prognostic methods by models and filtering methods.

In [64], a catalyst degradation model based on the physical deterioration mechanism is proposed by Zhang *et al*. This model is used to indicate the degradation rate of ECSA, and the area size can be indicated by the output voltage. Then the UKF is utilized to predict the RUL of a PEMFC system. The prediction results are located in 95 % confidence intervals when the prediction horizon is about 300 h. In practice, it is inconvenient to measure the ECSA, and the measurements may disturb the behavior of the fuel cell stack. The same semi-empirical voltage model with an adaptive unscented Kalman filter (AUKF) algorithm is utilized to realize the RUL prediction in [86]. An empirical aging model, adaptive Kalman filter, and nonlinear autoregressive exogenous (NARX) neural network are combined in [87] to realize the voltage prediction of Data Challenge in the short term.

Chen *et al*. proposed a hybrid method that combined three empirical voltage models (linear, logarithmic, and exponential) and UKF to realize the degradation prediction [88]. In the Mobypost project [89], ten fuel cell electric vehicles are developed for the postal delivery mission. There are 40 cells at each PEMFC, and the rated power and current are 1 kW and 34 A separately. The effectiveness of the proposed method is validated by the dynamic datasets of Mobypost. Results show that the logarithmic model has the best prediction performance in the short term (within 10 h).

Bressel *et al*. introduce the EKF to the long-term prognostic area of the PEMFC system [65], [90]-[96]. In Bressel *et al*.'s work, the semi-empirical voltage model is built based on the polarization curves of a single cell. The parameters of OCV, exchange current, overall resistance R_{over}, and limiting current i_L are calculated at each characterization time (168 h in this case) with the help of the Levenberg-Marquardt optimization method. Through analyzing these four parameters deviation, the $R_{over}(t) = R_{over0}(1 + \gamma(t))$ and $i_L(t) = i_{L0}(1- \gamma(t))$ change a lot, and the OCV and exchange current keep constant in the whole durability test. Where R_{over0} and i_{L0} are the initial value of R_{over} and i_L, and $\gamma(t)$ is the HI of "EKF-based degradation factor (DF)" [90]. Regarding the R_{over} and i_L as the state

variables, the EKF is used to estimate their linearity variation tendency.

Afterward, the same model of [90] is further verified in [91], [92], and [93]. The dataset in [91] is tested under the SS operating condition with characterization disturbances and voltage recovery. Several simulations based on the EKF are performed to analyze the sensitivity of the parameters affected by the initial value of R_{over} and i_L. Nevertheless, the datasets in [90] and [91] are under SS operating conditions. After that, the method is verified by the dynamic datasets in [92] and [65] separately. The datasets in [92] are tested under an automotive profile. The prediction horizon is about 30 h (175 hours' durability test) when the results are located into ± 10 % bound of real RUL. And the EKF extraction process is implemented in [65] under the dynamic micro-combined heat and power (μ-CHP) condition [97]. After the extraction of parameters by EKF, the model-based method of inverse first-order reliability method (IFORM) is used to extrapolate the degradation indicator until to the EoL threshold [65]. Besides, the RUL prediction results with a 90 % confidence interval are also presented, and the SH is about 200 h (1000 hours' durability test) when the results are located into ± 10 % error bound of the real RUL. Later, the RUL real-time estimation is validated on the hardware in the loop (HIL) platform [93].

Besides the semi-empirical voltage model of [90], a bond graph (BG) model of the PEMFC system is also proposed by Bressel *et al*. in [94] and [98]. Besides, the PF is used to replace EKF in [94], and the HI is renamed as "PF-based DF" in this paper. Comparison results under the SS (FC1), QD (FC2), and full dynamic operating conditions show that the BG-particle filter (BG-PF) hybrid method could have higher accuracy than [90]. Similar to the model in [94], an aging tolerant control method is realized in [95] and [96]. This method aims at generating the reference current and considering the degradation state at the same time. Besides, the reference value of the gas can be calculated by the energetic macroscopic representation (EMR) model, and the maximum power which can be provided by the fuel cell stack is identified by the maximum power point tracking (MPPT) algorithm [95], [96].

Recently, the FDKF and the four voltage degradation models (linear, quadratic, logarithmic, and exponential) are combined by Ao *et al*. [99]. This method is more efficient than the typical EKF because it can process the data in groups. Besides, the datasets of Data Challenge are used to verify its effectiveness in the long term.

Jouin *et al*. have worked on the application of PF in the prognostic area of the PEMFC system [61], [100]-[105], and [106]-[108] reviewed the current methods of RUL predictions. Three empirical models (linear, log-linear, and exponential) are used to represent the degradation rate of the PEMFC system in [61], and then the PF is taken to assist the prediction procedure. Based on the datasets of Data Challenge, the log-linear model shows the best performance in the long-term perspective with a 10 % error bound. When the prediction horizon is 500 hours, the prediction error is in the interval of [-25 h, +50 h].

Later, the prognostics which include characterization disturbances and voltage recovery are realized in [100], [101], and [102]. In [100] and [101], the aging model consists of three parts: 1) global power degradation is represented by a logarithmic and linear equation, 2) degradation's acceleration which is caused by the irreversible factors is represented by an exponential equation, and 3) the recovery is represented by an empirical equation. Then three paralleled PFs are used to deal with the models of each part separately, and the results are synchronized at the same time step. The prediction error is in the interval of [-28 h, +28 h] for a 950 hours' prediction horizon. To overcome the weakness of parameter initialization in [100] and further improve the prediction accuracy, an advanced empirical model is proposed in [102]. Compared to [100], the aging model in [102] consists of four parts, i.e., the degradation's acceleration is divided into two exponential equations. Four paralleled PFs are used to track the model,

and the prediction error is within 5 % when the prediction horizon is 500 h. The datasets in [100] and [102] are the same, and different PFs in the paralleling structure could improve the prediction accuracy. Nevertheless, recoveries due to the characterization are in the regular time intervals. These empirical models are built under SS or QD operating conditions.

Based on the typical loss model, both the time and current are used to build a semi-empirical model in [103]. Then the PF is utilized to realize the long-term predictions based on the dataset of FC1, and the prediction horizon is about 500 h. Similar to [103], another two semi-empirical models are proposed in [104] and [105] to describe the power behavior based on the dynamic dataset of transportation application and μ-CHP separately. Nevertheless, the degradation predictions in [104] and [105] are realized in the short-term perspective.

The adaptive particle filter (APF) is used in [57] to predict the stack voltage in the long term based on the datasets of Data Challenge [51]. Similar to [61], Kimotho *et al*. tested another two empirical models (pure logarithmic and polynomial model) apart from the linear, log-linear, and exponential model [57]. A self-healing factor is introduced to increase the adaptiveness, and its value is observed and updated after each characterization. The prediction error is in the interval of [-25 h, +25 h] (under 5 %) when the prediction horizon is about 500 h. Based on the datasets of Data Challenge, this method gets the highest score (0.7760) from the academic participants. Nevertheless, the self-healing factor should be determined in advance, and the setting process is largely based on empirical knowledge.

The recovery phenomena caused by the measurements of polarization curves and EIS make the prognostic difficult [109], [110]. Because the degradation rates of each two characteristic measurements are not the same, the sudden jumps phenomena would decrease the RUL prediction accuracy. The second-order polynomial model in [61] is used to describe the stack power, Zhang *et al*. also proposed other three functions to model the recovery state. Moreover, the parameters in the recovery model are extracted from the ECM of the EIS curves. Finally, with the help of PF, the RUL is predicted in the long-term (about 500 h) with the datasets of Data Challenge.

Zhou *et al*. proposed a multi-physical (electrical, fluidic, and thermal domain) aging model in [111]. And then the PF is used to predict the degradation in the long-term perspective, and the prediction horizon is 150 h and 250 h. Finally, 3 datasets (12 A/30 ℃, 30 A/35 ℃, and 44 A/40 ℃) from the Ballard NEXA fuel cell stack are used to verify the robustness of this method.

Similar to the degradation-related parameter reconstruction process of [94], the static HI is extracted from the modeling of polarization curves. Then the PF is used in [112] to predict the RUL in the long term, and the data of FC1 from Data Challenge are used for the validation. The classifications of model-based methods (models with filtering approaches) are shown in Tab. 3 and Tab. 4.

Tab. 3 The classification of prognostic methods by models and KF.

Year	Reference	Method	Health indicator	Operating state	PH
2021	[99]	Four empirical models + FDKF	Voltage	SS and QD	≤ 500 h
2019	[88]	Three empirical models + UKF	Voltage	Dynamic	≤ 10 h
2017	[86]	Semi-empirical model + AUKF	Voltage	SS	Single-step

2016	[91]	Semi-empirical model + EKF	EKF-based DF	SS	≤ 1500 h
2016	[92]	Semi-empirical model + EKF +IFORM	EKF-based DF	Dynamic	≤ 120 h
2016	[65]	Semi-empirical model + EKF +IFORM	EKF-based DF	Dynamic	≤ 800 h
2015	[90]	Semi-empirical model + EKF	EKF-based DF	SS	≤ 800 h
2012	[64]	Catalyst degradation model + UKF	Voltage	Dynamic	≤ 300 h

Tab. 4 The classification of prognostic methods by models and PF.

Year	Reference	Method	Health indicator	Operating state	PH
2021	[112]	Semi-empirical model + PF	PF-based DF	SS	≤ 500 h
2018	[109]	Four empirical models + PF	Power	SS and QD	500 h
2017	[111]	Multi-physical model + PF	Voltage	SS	≤ 250 h
2016	[110]	Four empirical models + PF	Power	SS and QD	500 h
2016	[102]	Four empirical models + paralleled PFs	Power	SS	≤ 500 h
2016	[94]	Semi-empirical model + PF	PF-based DF	SS and dynamic	≤ 900 h
2016	[104]	Semi-empirical model + PF	Power	Dynamic	Single-step
2015	[103]	Semi-empirical model + PF	Voltage	SS	≤ 500 h
2015	[105]	Semi-empirical model + PF	Power	Dynamic	Single-step
2014	[61]	Three empirical models + PF	Voltage	SS and QD	≤ 850 h
2014	[100], [101]	Three empirical models + paralleled PFs	Power	SS and QD	≤ 950 h
2014	[57]	Five empirical models + APF	Voltage	SS and QD	500 h

3.1.4. Models with ML methods

The prognostic methods of models and ML are classified in Fig. 7, including majorly artificial neural network (ANN), Gaussian process (GP), and support vector machine (SVM) methods. Besides, the models can also be combined with the time-frequency analysis method of wavelet transform (WT).

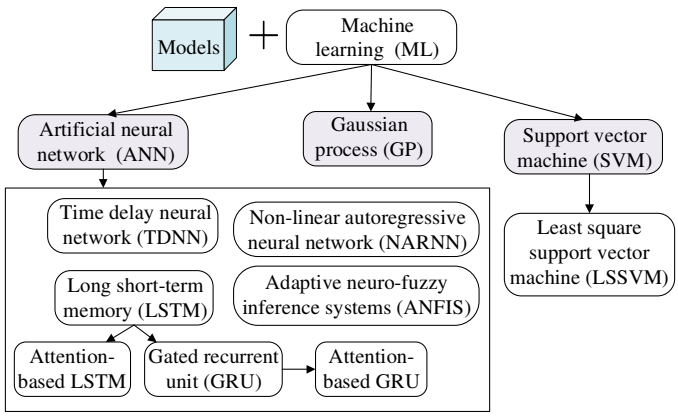


Fig. 7. The prognostic methods of models and ML.

The discrete wavelet transform (DWT) and two degradation models, i.e., polynomial regression and autoregressive integrated moving average (ARIMA) models are used to realize the online RUL prediction [113]. The data in one week is compressed by DWT by the scale factor of 2^{k+1} (decomposition level k = 2), and then the models are utilized to predict the coefficients of the next week. Finally, the output voltages of the next week are reconstructed by the inversed discrete wavelet transform (IDWT). Results of FC1 and FC2 show that the data in 168 h can be shortened into 21 h, and fewer data are needed in the models to realize the long-term RUL prediction (PH = 168 h).

To improve the prediction accuracy of [113], the data-driven method of non-linear autoregressive neural network (NARNN) and an empirical fuel cell voltage model are combined to realize the degradation prediction of the PEMFC system [114]. The moving window method is used to improve the prediction accuracy, and the weights of these two methods are updated in each step (time interval is 168 h). Finally, this hybrid method is validated based on the datasets of Ballard NEXA stack (36 A/40 ℃), proton motor 200 (PM200) fuel cell stack (64 A/58 ℃), and Data Challenge. Based on the same datasets of FC1 and FC2, the prediction accuracy of [114] is higher than [113] in terms of RMSE. Afterward, Zhou *et al*. proposed another hybrid method that combined the autoregressive and moving average (ARMA) model and time delay neural network (TDNN) [115]. Before executing this hybrid method, a PF-based empirical model and physical aging model (PAM) are utilized to remove the non-stationary trend of the original stack voltage. The linear part is filtered by the ARMA model, and the nonlinear part is processed by the TDNN. This ARMA-TDNN method can handle the voltage signal in different patterns, and the improvements are validated by 3 datasets of Ballard NEXA stack (12 A/30 ℃, 30 A/35 ℃, and 44 A/40 ℃) and 2 datasets of Data Challenge.

To overcome the weaknesses of typical PF [115], the regularized particle filter (RPF) which is based on three state models (exponential, log-linear, and linear) is proposed by Cheng *et al*. [116]. The hybrid approach is combined by the RPF and data-driven method of the least square support vector machine (LSSVM). The output voltage is first predicted by LSSVM, and then the predicted value is used in the RPF structure. Besides, the confidence interval (90 %) and PDF curve can be given by LSSVM-RPF, which are helpful for the users to make proper decisions. Based on the datasets of Data Challenge, the prediction performance of LSSVM-RPF is better than typical PF and RPF. Nevertheless, both in [115] and [116], the model-based method is realized in the long term, and the data-driven method is realized in the short term.

Apart from the PF, the GP can also present the degradation tend in a probability distribution form [117]. An empirical aging model is developed by Zhu *et al*., and the GP is used to predict the stack voltage under the SS operating condition (FC1). During the prediction, the GP model is updated every 50 h based on the new measured data.

Thus, the prediction horizon of GP is 50 h. Recently, the GP is also used to realize the degradation prediction by Xie *et al*. [118] and Deng *et al*. [119]. And the single-step ahead voltage is predicted by the datasets of FC1 and FC2.

Another hybrid prognostics method is proposed by Liu *et al*. [120]. The semi-empirical voltage degradation model, AUKF, adaptive neuro-fuzzy inference systems (ANFIS), and particle swarm optimization (PSO) are combined to realize the long-term RUL prediction.

Yang *et al*. constructed a hybrid prognostic method in [121] and [122] by combining empirical models, PF, and long short-term memory (LSTM) network. The parameters of three models (linear, logarithmic, and exponential) are learned by PF, and the RUL is estimated in the long term by the LSTM network.

The LSTM and autoregressive integrated moving average (LSTM-ARIMA) method is proposed in [123]. The LSTM-ARIMA structure can decrease the influences of overfitting and recovery phenomenon, and it is tested in the short term. The methods in [117]-[123] are verified by the offline datasets of Data Challenge.

Ma *et al*. proposed a hybrid prognostic method based on the EKF and LSTM network [124]. The internal parameters of limiting current, exchange current, and overall resistance is extracted by a PAM. Then the EKF is utilized to estimate the three parameters based on the output voltage. Finally, the data-driven method of LSTM is used to realize the voltage (or virtual steady-state voltage) prediction in the long term under different operating conditions.

Based on the work of Zhou *et al.* [111], the output voltage of a single cell is built by Zuo *et al.* [125]. And four types of deep learning methods i.e., LSTM, gated recurrent unit (GRU), attention-based LSTM, and attention-based GRU are compared in the short-term prediction of voltage.

Based on the degradation-related parameter extraction process of [65] and the multi-step ahead echo state network (ESN) prediction structure of [156], the long-term degradation prediction under the dynamic operating condition is realized by Yue *et al.* [126]. The dynamic HI is the same with [65], and the dataset from [156] is used for the validation.

Recently, a symbolic-based LSTM prognostics model is proposed by Wang *et al*. [127]. The reconstruction process of dynamic HI is similar to the work in [66] and [156]. The historical data (stack temperature, current, and voltage) are separated into different segments by the sliding data window (at hourly level), and then they are used for extracting the HI. A modified degradation behavior model is built based on the polarization curve, and different kinds of resistance in the fuel cell are modeled in detail. Then the LSTM is used to predict the HI trend in the long term, and the PH is about 700 h in a 1262 hours' durability test. The classification of model-based methods (models with ML methods) is shown in Tab. 5.

Tab. 5 Classification of prognostic methods of models and ML.

Year	Reference	Methods	Health indicator	Operating state	PH
2021	[124]	Physical model + EKF+ LSTM	Voltage + EKF-based DF	SS, QD, and Dynamic	≤ 450 h
2021	[125]	Empirical models + some deep learning methods	Voltage	SS and QD	Single-step
2021	[126]	Semi-empirical model + ESN	EKF-based DF	Dynamic	≤ 900 h
2021	[127]	Semi-empirical model + LSTM	Hybrid HI	Dynamic	≤ 688 h
2019	[120]	Semi-empirical model + AUKF +ANFIS	Voltage	SS and QD	≤ 1000 h

2019	[123]	ARIMA + LSTM	Voltage	SS and QD	Single-step
2018	[115]	(Empirical and physical model) + (ARMA and TDNN)	Voltage	SS	Single-step
2018	[116]	Three empirical models + RPF + LSSVM	Voltage	SS	Single-step
2018	[117]	Empirical model + GP	Voltage	SS	50 h
2017	[114]	Empirical model + NARNN	Voltage	SS and QD	168 h
2017	[121], [122]	Three empirical models + PF + LSTM	Voltage	SS and QD	400 h

3.2 Data-driven methods

From the literature known so far, the data-driven methods are mainly based on the ML framework, and many kinds of ML approaches have been used in the RUL predictions of the PEMFC system. The types of data-driven methods are summarized in Fig. 8.

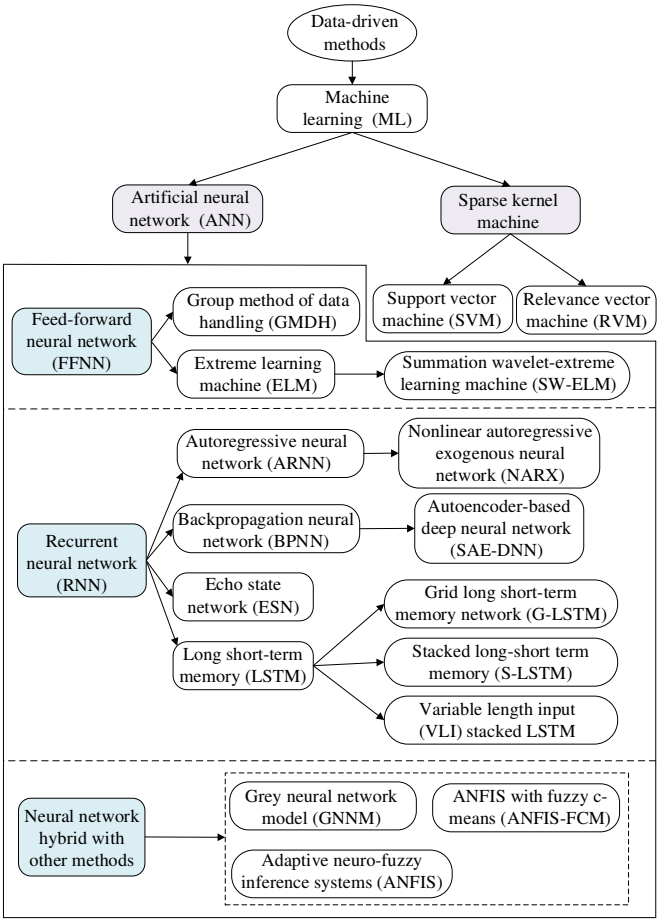


Fig. 8. The types of data-driven methods.

In [128], the temporal variation of stack voltage is learned by ANFIS, and then the new stack voltages are predicted by the training structure. To improve the prediction accuracy, Silva *et al*. divided the voltages into the normal operation part and external perturbations part (e.g., transient process), and only the normal operation part is used to train the ANFIS structure. This splitting approach can efficiently improve the prediction behaviors of two PEMFC stacks. The prediction horizon of these two long-term tests is about 500 h under the SS and QD operating conditions of Data Challenge. The long-term prediction is realized by the iterative approach and single-step ahead prediction process.

The parameters are set by the trial-and-error process which relies heavily on the experience. A simple comparative study of ANFIS, feed-forward neural network (FFNN), and autoregressive neural network (ARNN) is presented in [129]. The ANFIS with fuzzy c-means (ANFIS-FCM) method is proposed by Liu *et al.* [130] to realize the short-term prediction of stack voltage, and the parameters of ANFIS-FCM are optimized by PSO to improve the adjustment efficiency. Besides, the WT is also combined with ANFIS-FCM to obtain a better performance. A deep belief network and extreme learning machine are used to realize the short-term prediction of voltage [131]. The datasets in [128], [130], and [131] come from Data Challenge.

The summation wavelet-extreme learning machine (SW-ELM) is introduced to the PEMFC system by Javed *et al.*, and it is fully explored at a long-term prediction scale under SS operating conditions [52], [53], [132]. The SW-ELM can be regarded as the combination of ANN and WT theory. Two activation functions are used for each hidden node of ANN, and two parameters' initialization (weights and bias) can decrease the effects of uncertain perturbations. Therefore, its ability to deal with nonlinearity and disturbance would be enhanced. Long-term prediction results with different training lengths have been researched in [52], and the prediction horizon is about 500 h. Nevertheless, the single prediction has some limitations in dealing with the uncertainties of degradation, disturbances of the operation conditions, parameter sensitivity, etc. To overcome the weakness of a single prediction, the SW-ELM with an ensemble of constraints is proposed in [53] and [132] to improve the prediction accuracy and robustness. One hundred predictions by SW-ELM with different learning parameters at each training length are implemented, and then the mean RUL and median RUL of all predictions are calculated and compared. Both in [53] and [132], the training length starts from 850 h (half-length of the data), the prediction interval is 50 h, and the prediction horizon is from 50 h to 850 h. Besides, the data in [52] is also tested by the ensemble of constraints-based SW-ELM approach in [132]. In this dataset, training length starts from 600 h, and the prediction horizon is less than 550 h. Results show that the median RUL is more interesting because the prediction of them usually belongs to the underestimations. The prognostic with ensemble structure can minimize the error probability than the individual prediction. Besides the SS operating condition, the ensemble SW-ELM approach is also tested by Javed *et al.* under the variable load [133]. The results of SW-ELM with or without incremental learning are compared under the dynamic durability tests for μ-CHP. In [133], the new measurement data are used for training when they are available, and the prediction is realized in the single-step ahead process in essence.

Based on the ensemble structure and WT principle [133], the approaches of group method of data handling (GMDH) network [134], extreme learning machine (ELM) [135], [136], NARX neural network [137], and ESN [138] are used to deal with the multi-timescale features of PEMFC system. Firstly, the original signal is decomposed into several sub-waveforms by the WT. Secondly, the sub-waveforms are predicted by different data-driven methods separately. Finally, an ensemble structure is utilized to integrate the multi-timescale features and improve the prediction accuracy. In [134], the original stack voltages are decomposed into 3 layers of FC1 and 4 layers of FC2. The stack voltages do not need to be normalized in the training process, and the short-term prediction is realized from a single-step ahead to 5-step ahead (about 5 h) process. Compared to [134], the GMDH is replaced by the enhanced ELM method in [135], and the self-adaptive differential evolutionary (SaDE) method is used to optimize the parameters of ELM. The prediction procedure of [135] is also similar to [53], and one hundred predictions are implemented to weaken the randomness of ELM. Besides, the influence of different training lengths is also analyzed with the same datasets of [134]. Later, the WT and ELM are used in [136] to realize the degradation prediction, and 3

decomposition layers are chosen for the dynamic datasets of MobyPost. Different from [135], the genetic algorithm (GA) is used to optimize the parameters of ELM, and a single step ahead of stack voltage prediction is realized under the dynamic operating condition [136]. Recently, the WT and ESN are combined in [138] to realize the long-term RUL prediction under 3 dynamic tests of μ-CHP. Nevertheless, the setting principle of the decomposition layer is unclear, and the physical meaning of the features in different layers needs to be further explored [134]-[138].

Also based on the datasets of FC1 and FC2, the grey neural network model (GNNM) method is proposed by Chen *et al.* to realize the single-step ahead voltage prediction [139]. The PSO is used to optimize the parameters of GNNM, and the moving window method is used to improve the prediction accuracy. Later, the backpropagation neural network (BPNN) is applied to construct the relationship of inputs (current, humidity, temperature, etc.) and output (voltage), and it is tested by the datasets of FC1, FC2, and MobyPost [140]. Apart from the GA [136] and PSO [139], the mind evolutionary algorithm is also used in [140] to optimize the initial weights and neuron numbers of BPNN. Besides, the BPNN with different activation functions are tested, and the voltage prediction results are compared with the traditional relevance vector machine (RVM) and SVM. The BPNN, LSTM, and GRU are compared in [141]. Nevertheless, in [134]-[137], [139]-[141], the degradation prediction of the stack voltage is all realized in the short term.

The modified RVM and self-adaptive RVM are put forward to predict the degradation of the PEMFC system by Wu *et al.* in [142] and [143] separately. Compared with the classical SVM, the RVM has higher selecting freedom in kernel functions, and fewer vectors are needed in RVM which can further decrease the computational complexity. Moreover, the RVM can realize the predictions with confidence intervals, i.e., it has the probabilistic prediction ability which is more interesting in practice than the single point prediction. Two experimental datasets (400 hours' durability testing) from the Ballard NEXA fuel cell stack under the SS operating conditions (30 A/ 35℃ and 36 A/40 ℃) are used to validate the Gaussian kernel-based RVM in [142]. The output voltage is used to indicate the degradation state. Prediction results in different training lengths are discussed, and the prediction horizon is 50 h, 100 h, and 160 h separately. Besides, the modified RVM has shown a better performance than SVM, and the confidence interval of 99.7 % is also given in the prediction results. Later, the self-adaptive RVM is proposed in [143] to improve the kernel width selection and prediction accuracy. Besides the 2 datasets in [142], the other 3 datasets are used to validate the prognostic performance, i.e., two tests (12 A/30 ℃ and 44 A/40 ℃) are from the Ballard NEXA fuel cell stack under the SS operating conditions, and the last test (30–97 A/58 ℃) is from PM200 fuel cell stack under the dynamic operating condition. The short-term predictions are realized based on the multi-step ahead prediction process, i.e., from single-step ahead (4 h) to 5-step ahead (20 h). The prediction performance is improved when compared with the classical SVM and original RVM. Nevertheless, the effectiveness of RVM in the long-term timescale needs to be further researched especially under dynamic operating conditions.

LSTM network is an improved structure of the recurrent neural network (RNN) [144]-[146], and it can avoid the issues of gradient exploding and vanishing. The LSTM method is introduced to the degradation prediction of the PEMFC system by Ma. *et al.* [144]. Apart from the 4 datasets (12 A/30 ℃, 30 A/35 ℃, 36 A/40 ℃, and 44 A/40 ℃) from the Ballard NEXA fuel cell stack in [143], another dataset with 1 hour time interval from PM200 fuel cell stack (64 A/58 ℃) is used to validate the effectiveness of LSTM network in [144]. Later, the grid long short-term memory network (G-LSTM) is proposed in [145] to further improve the prediction accuracy of output voltage. Besides the 5 datasets from [144], the other 2 datasets from Data Challenge, and 1 dynamic dataset (20–99 A/58 ℃) from the

PM200 fuel cell stack (time interval is 4 h) are used to test the G-LSTM network in [145]. Moreover, the prediction accuracy is guaranteed by the updated sliding window size. Similar to the works in [144] and [145], the LSTM network is used in [146] to predict the output voltage of 2 datasets from Data Challenge. After dealing with the locally weighted scatterplot smoothing, the voltages are resampled at the time interval of 1 hour. Results show that the predictions by the LSTM network are more accurate than the BPNN both in FC1 and FC2. The stacked long-short term memory (S-LSTM) is proposed in [147], the bi-directional LSTM is proposed in [148] and the 2 datasets from Data Challenge are used to verify their effectiveness. The variable length input (VLI) stacked LSTM is proposed in [58] to realize the online voltage prediction, and this method is verified on the SS operating conditions. The PF and LSTM are combined to improve the long-term prediction accuracy in [149]. A navigation sequence driven LSTM (NSD-LSTM) for long-term prediction is proposed in [150].

Later, Liu. e*t al*. used the method of sparse autoencoder-based deep neural network (SAE-DNN) to predict the output power of FC2 [54]. The single-step ahead prediction is realized after the filtering and resampling of the stack power. Compared to K-nearest neighbor (KNN) and SVM regression algorithms, the SAE-DNN can reach a higher accuracy at different training lengths. Nevertheless, the output voltage/power predictions in [54], [58], [144]-[148] are realized by the single-step ahead prediction process. The prediction horizon in these papers depends on the time intervals of the resampled data, i.e., the 1-hour or 4-hour ahead prediction is realized in these datasets.

The ESN is an improved design and training architecture of RNN. Compared to the RNN, the hidden layer is replaced by a large randomly generated reservoir which reflects the dynamic topologies of the neurons. The ESN can overcome the shortcomings of traditional ESN such as heavy computational burden, slow convergence rate, and gradient exploding. The basic ESN structure is introduced to the prognostic area of the PEMFC system by Morando *et al*. [55], [151]. It is used to predict the mean voltage of cells in the short term, and two prediction structures (direct approach and parallel approach) have been fully explored. The advantages of ESN and its parameter configurations are presented in [152], and the results of direct structure (in the short term) and iterative structure (in the long term) are compared. The double ESN structure is used after the wavelet filtering to improve the prediction accuracy in [153], and the parameter settings are the same as [152]. Based on the results of the analysis of variance (ANOVA) method in [154], the multi-reservoir ESN structure is developed in [155] to evaluate the mean voltage of cells in the long term. And this structure can avoid the parameters optimization process by setting the reservoirs with different spectral radii. The voltage is used as the HI in the above articles [55], [151]-[155], and the predictions are implemented in the single-step ahead or multi-step ahead under the SS and QD operating conditions.

After proposing the dynamic HI in [66], an ensemble ESN structure is proposed by Li *et al.* [156]. This structure is used to realize the long-term RUL prediction under the SS and dynamic operating conditions. The multi-step ahead prediction is realized in [156] by the iterative principle. This method is tested at different training lengths, and the acceptable prognostic can be achieved when the prediction horizon is within 350 h under both the SS and dynamic operating conditions.

Different ESN prediction structures are compared by Hua *et al*. [157]. The single-step ahead (in the short-term of 0.5 h) and multi-step ahead (in the long-term of 500 h) prediction structures are discussed based on the 2 datasets of Data Challenge. Results show that the ESN with fixed output weight matrix has a lower computing burden, and the ESN with an updated output weight matrix has a higher prediction accuracy. The cycle reservoir with jump ESN (CRJ-ESN) is used by Jin *et al*. [158]. This structure could change the connection mode of neurons in the reservoir

and speeds up the linear fitting process. After resampling the data by an interval of 700 data points (≈ 6 h), this CRJ-ESN structure is tested by the 2 datasets of Data Challenge in the short term. Based on the same datasets of [158], the stacked ESN and GA method are combined to realize the single-step ahead prediction in [159].

Recently, the ESN and Markov chains are used to predict the degradation under a variable load profile (repeated load cycle) [160], and this method can obtain satisfying results without a great effort for data processing. Nevertheless, the HI of [160] is stack voltage, and the prediction is also realized by the single-step ahead process.

The data-driven approach of discrete wavelet transform-echo state network-genetic algorithm (DWT-ESN-GA) is proposed to improve the RUL prediction performance [161]. Based on the "compression-prediction-reconstruction" technique, degradation characteristics in 2016 data points can be effectively represented by only 253 data points. Compared to the model-based methods in [113] and [114], the prediction accuracy of DWT-ESN-GA is improved.

Based on the multi-input and multi-output ESN (MIMO-ESN) structure, the operating parameters can be regarded as the inputs together with the HI to improve the prediction accuracy [56], [67], [162], and [163]. Experimental results show that the MIMO-ESN can improve the prediction accuracy in different operating conditions. In [56], the operating parameters, such as stack current, stack temperature, and the pressures of reactants are utilized as other inputs of ESN. The feasibility and effectiveness of the MIMO-ESN are verified on the 2 datasets (FC1 and FC2) of Data Challenge. The PH of voltage is about 500 h. Later, a dynamic HI of RPLR is proposed in [67], and the effects of scheduled current are quantitatively analyzed based on 3 dynamic datasets of μ-CHP. The PH of voltage is lower than 600 h.

Similar to the structure of [56], a multiple-input ESN is also presented in [162] and [163]. The measured ambient temperature and the predicted stack voltage are used as the double inputs to predict the RUL recently [162], and the influence of the temperature is analyzed by the dynamic dataset of Mobypost. The PH of voltage is about 4000 h. The least absolute shrinkage and selection operator-echo state network (LASSO-ESN) is proposed in [163]. The input parameters of ESN are evaluated and selected by LASSO, the less degradation-related parameters are eliminated, and the more degradation-related parameters are used for the RUL prediction. Results on the 2 datasets of Data Challenge also show that the optimal input parameters are different under various operating conditions, and the stack current is an interesting variable [162]. The PH of voltage is about 500 h. Nevertheless, a major potential assumption of multiple inputs ESN is that the operating parameters are measurable, schedulable, or programmable. Besides, in this prediction structure, the operating parameters are measured and used as the inputs of ESN, and the HI is predicted in the long term by the iterative principle. The classification of data-driven methods is shown in Tab. 6.

Tab. 6 Classification of data-driven methods.

Year	Reference	Methods	Health indicator	Operating state	PH
2021	[137]	DWT-NARX	Voltage	Dynamic	Single-step
2021	[158]	CRJ-ESN	Voltage	SS and QD	Single-step
2021	[160]	Ensemble ESN	Voltage	Dynamic	Single-step
2021	[161]	DWT-ESN-GA	Power and RPLR	SS, QD and dynamic	168 h
2021	[67]	Multi-input ESN	RPLR	Dynamic	≤ 600 h
2021	[163]	Multi-input ESN	Voltage	SS and QD	≤ 500 h
2020	[140]	Mind evolutionary	Voltage	SS	Single-step

		algorithm-BPNN			
2020	[147]	S-LSTM	Voltage	SS and QD	1 h
2020	[58]	VLI-S-LSTM	Voltage	SS	Single-step
2020	[157]	ESN	Voltage	SS and QD	0.5 h and ≤ 500 h
2020	[56]	Multi-input ESN	Voltage	SS and QD	≤ 500 h
2020	[162]	Multi-input ESN	Voltage	Dynamic	≤ 4000 h
2019	[130]	ANFIS-FCM	Voltage	SS and QD	Single-step
2019	[135]	DWT-ELM	Voltage	SS	1 h
2019	[136]	DWT-ELM	Voltage	Dynamic	Single-step
2019	[139]	PSO-GNNM	Voltage	SS, QD and dynamic	5 mins
2019	[146]	LSTM	Voltage	SS and QD	1 h
2019	[54]	SAE-DNN	Voltage	QD	1 h
2019	[156]	Ensemble ESN	Virtual voltage	SS and Dynamic	≤ 750 h
2018	[144]	LSTM	Voltage	SS	4 h
2018	[145]	Grid-LSTM	Voltage	SS, QD and dynamic	1h and 4 h
2018	[155]	Multi-reservoir ESN	Voltage	SS	≤ 900 h
2017	[134]	DWT-GMDH	Voltage	SS and QD	≤ 5 h
2016	[132]	Ensemble SW-ELM	Voltage	SS	≤ 850 h
2016	[133]	SW-ELM	Power	Dynamic	Single-step
2016	[142]	Modified RVM	Voltage	SS	≤ 160 h
2016	[143]	Self-adaptive RVM	Voltage	SS and dynamic	≤ 20 h
2016	[153]	Double ESN	Voltage	SS	1400 h
2015	[52]	SW-ELM	Voltage	SS	≤ 500 h
2015	[53]	Ensemble SW-ELM	Voltage	SS	≤ 850 h
2014	[152]	ESN	Voltage	SS	1400 h
2014	[128]	ANFIS	Voltage	SS and QD	500 h
2013	[55]	ESN	Voltage	SS	30 s

4. Conclusion and perspectives

Firstly, the background, working principles, the degradations caused by mechanical, chemical, and operating conditions of the PEMFC system are briefly introduced. Secondly, the key processes of PHM are given in detail. Different types of data processing techniques, HIs, prognostic methods, and performance evaluation metrics are presented. Meanwhile, the definitions of remaining useful life, end of life, prediction steps, prediction horizon, and satisfactory horizon are stressed. Finally, different kinds of static and dynamic models of PEMFC for the prognostic are given, and a comparative study on degradation prediction methods, i.e., model-based and data-driven, are presented.

From our point of view, some aspects need to be further developed on the degradation prediction of the PEMFC system.

1) Find more and more convenient and efficient dynamic HIs. The degradation state of the PEMFC stack can be reflected by the HIs, and a proper HI is important for the prognostic methods. In the dynamic operating conditions, the monotonic decreasing tendency of static HI (e.g., power, voltage, geodesic distance, and other hybrid forms)

becomes less obvious because of the uncertainty change of load current. Thus, the traditional static HIs cannot be used anymore. In addition to the dynamic HIs of "electrochemical catalytic surface area", "extended Kalman filter-based degradation factor", "virtual steady-state voltages", and "relative power-loss rate", other dynamic HIs should be explored in the practical condition. A good HI in the PEMFC system should have the properties of low computational complexity, can be implemented online, has generally monotonic trend along with time, can be used for all working states, etc.

2) Realize the degradation prediction on the mid-term or long-term scale. The lifetime of the PEMFC system is in hundreds or thousands of hours, the short-term prediction (PH $\leq$ 24 h in this paper) cannot provide enough time for the users to take some maintenance actions in advance. Thus, the short-term prediction does not have much practical meaning at all if the goal is to extend the lifespan even though the prediction error seems small. The degradation predictions in the middle term (24 h $<$ PH $\leq$ 168 h) or long term (PH $>$ 168 h) are more useful for the operators because the lifetime can be extended by regulating the operating conditions.

3) Improve the degradation prediction performance. Improving the prediction performance, including but not limited to accuracy, efficiency, and robustness. The stochastic disturbances produced by the sensor noises during the measurements, unknown disturbances from the environment and operating parameters, load irregular varying caused by user habits, and the uncertainties of the prognostic methods themselves all would affect the prediction performance.

Quantitative analysis of the above-mentioned uncertainties, accurately extracting the degradation-related information, exploring the anti-disturbance methods can enhance the prediction performance. Different fault signals would also be measured along with the degradation data. Using the diagnostic technique to separate the fault signal in advance helps improve the prediction accuracy. The ensemble structure of the prediction methods can be used to improve their robustness and accuracy from the probability point of view. The exploration of the methods which can provide the confidence intervals and probability density function would also be welcomed. Further exploring the prediction efficiency improvement methods (i.e., decreasing the volume of data) is necessary because of the online requirement.

4) Realize the online prediction of RUL by exploring different prognostic methods. The online prediction has more practical meanings than offline, and there are only a few articles that have initiated the related works as far as the author knows. The common idea of online RUL prediction is building the mathematical models or "black box" relationships by the historical data, and then realizing the mid-term or long-term prediction in a very short time. The wavelet-based approach and some models (polynomial regression and ARIMA) have been combined to realize the online prediction by some researchers. Nevertheless, for the data-driven method, the complex structure of machine learning methods makes the implementation of the online process more difficult. The optimization of multiple hyperparameters has also limited the execution time of the algorithm, and using some online parameter optimization methods would be helpful. Thus, how to realize the online RUL prediction efficiently is worthy of study.

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References

[1] United Nations Department of Economic and Social Affairs (UN DESA), "World Population Prospects 2019." Available online: https://population.un.org/wpp/

[2] Economics and Technology Research Institute (ETRI) of China National Petroleum Corporation (CNPC), "2050 World and China energy outlook, " 2019. https://pdf.dfcfw.com/pdf/H3_AP201912151371851724_1.pdf?1576501894000.pdf

[3] M. Yue, H. Lambert, E. Pahon, R. Roche, S. Jemei, and D. Hissel, "Hydrogen energy systems: A critical review of technologies, applications, trends and challenges," *Renew. Sustain. Energy Rev.,* vol. 146, p. 111180, Aug. 2021.

[4] M., Shawna, S. Joseph, S. Sunita, M. Eric, S. Ned, P. Dimitrios, *et al.*, "Department of Energy Hydrogen Program Plan," United States Department of Energy (U.S. DOE), Washington DC, United States, Tech. Rep., no. DOE/EE-21288387, Nov. 2021.

[5] Hydrogen and Fuel Cell Technologies Office (HFTO) of U.S. DOE, "Hydrogen storage." Available online: https://www.energy.gov/eere/fuelcells/hydrogen-storage

[6] Q. Li, T. Wang, S. Li, W. Chen, H. Liu, E. Breaz, et al., "Online extremum seeking-based optimized energy management strategy for hybrid electric tram considering fuel cell degradation," *Appl. Energy*, vol. 285, p. 116505, Jan. 2021.

[7] Q. Li, L. Yin, H. Yang, T. Wang, Y. Qiu, and W. Chen, "Multiobjective optimization and data-driven constraint adaptive predictive control for efficient and stable operation of PEMFC system," *IEEE Trans. Ind. Electron.*, vol. 68, no. 12, pp. 12418-12429, 2021.

[8] T. Wang, Q. Li, H. Yang, L. Yin, X. Wang, Y. Qiu, *et al.*, "Adaptive current distribution method for parallel-connected PEMFC generation system considering performance consistency," *Energy Convers. Manage.*, vol. 196, pp. 866-877, Sep. 2019.

[9] Y. Wang, B. Seo, B. Wang, N. Zamel, K. Jiao, and X. C. Adroher, "Fundamentals, materials, and machine learning of polymer electrolyte membrane fuel cell technology," *Energy and AI*, vol. 1, p. 100014, Aug. 2020.

[10] M.-C. Péra, D. Hissel, H. Gualous, and C. Turpin, "Electrochemical components," Hoboken, New Jersey, United States: John Wiley & Sons, ch. 3, pp. 151–207, Jul. 2013.

[11] T. Wang, Q. Li, X. Wang, Y. Qiu, M. Liu, X. Meng, et al., "An optimized energy management strategy for fuel cell hybrid power system based on maximum efficiency range identification," *J. Power Sources*, vol. 445, p. 227333, Jan. 2020.

[12] X. Gu, Z. Hou, and J. Cai, "Data-based flooding fault diagnosis of proton exchange membrane fuel cell systems using LSTM networks," *Energy and AI*, vol. 4, p. 100056, 2021.

[13] Z. Hua, Z. Zheng, F. Gao, and M.-C. Péra, "Challenges of the remaining useful life prediction for proton exchange membrane fuel cells," *in Proc. IECON 45th Annu. Conf. IEEE Ind. Electron. Soc.*, Lisbon, Portugal, pp. 6382-6387, Oct. 2019.

[14] Q. Li, X. Meng, F. Gao, G. Zhang, and W. Chen, "Approximate cost-optimal energy management of hydrogen electric multiple unit trains using double Q-learning algorithm," *IEEE Trans. Ind. Electron.*, pp. 1-1, 2021.

[15] Q. Zhang, C. Harms, J. Mitzel, P. Gazdzicki, and K. A. Friedrich, "The challenges in reliable determination of degradation rates and lifetime in polymer electrolyte membrane fuel cells," *Curr. Opin. Electrochem.*, vol. 31, p. 100863, Feb. 2022.

[16] A. Kongkanand, D. Masten, C. Wang, J. Chen, J. Spendelow, B. James, *et al.*, "Automotive fuel cell targets and status," U.S. DOE, Washington DC, United States, Tech. Rep., no. 20005, Aug. 2020.

[17] E. J. F. Dickinson and G. Smith, "Modelling the proton-conductive membrane in practical polymer electrolyte membrane fuel cell (PEMFC) simulation: A review," *Membranes*, vol. 10, no. 11, p. 310, Oct. 2020.

[18] R. O'hayre, S.-W. Cha, W. Colella, and F. B. Prinz, "Fuel cell fundamentals," 3rd ed., Hoboken, New Jersey, United States: John Wiley & Sons, Apr. 2016.

[19] E. L. Miller, S. T. Thompson, K. Randolph, Z. Hulvey, N. Rustagi, and S. Satyapal, "US Department of Energy hydrogen and fuel cell technologies perspectives, " *MRS Bulletin*, vol. 45, no. 1,, pp. 57–64, Sep. 2020.

[20] Y. Zhou, A. Ravey, and M.-C. Péra, "A survey on driving prediction techniques for predictive energy management of plug-in hybrid electric vehicles," *J. Power Sources*, vol. 412, pp. 480-495, Feb. 2019.

[21] H. Q. Nguyen and B. Shabani, "Proton exchange membrane fuel cells heat recovery opportunities for combined heating/cooling and power applications," *Energy Convers. Manage.,* vol. 204, p. 112328, Jan. 2020.

[22] D. Hart, S. Jones, and J. Lewis, "The fuel cell industry review 2020," E4tech, Lausanne, Switzerland, Mar. 2021. https://fuelcellindustryreview.com/

[23] P. Pei and H. Chen, "Main factors affecting the lifetime of proton exchange membrane fuel cells in vehicle applications: A review," *Appl. Energy*., vol. 125, pp. 60-75, Jul. 2014.

[24] A. G. Olabi, T. Wilberforce, and M. A. Abdelkareem, "Fuel cell application in the automotive industry and future perspective," *Energy*, vol. 214, p. 118955, Jan. 2021.

[25] M. Mohammadi Taghiabadi and M. Zhiani, "Degradation analysis of dead-ended anode PEM fuel cell at the low and high thermal and pressure conditions," *Int. J. Hydrog. Energ.*, vol. 44, no. 10, pp. 4985-4995, Feb. 2019.

[26] B. James, J. Huya-Kouadio, C. Houchinsand, R. Ahluwalia, M. Ulsh, D. Masten, *et al.*, "Durability-adjusted fuel cell system cost," U.S. DOE, Washington DC, United States, Tech. Rep., no. 21001, Jan. 2021.

[27] A. Wilson, G. Kleen, and D. Papageorgopoulos, "Fuel cell system cost - 2017," U.S. DOE, Washington DC, United States, Tech. Rep., no. 17007, Sep. 2017.
[28] Hydrogen and Fuel Cell Technologies Office of U.S. DOE, "Comparison of fuel cell technologie", Washington DC, United States. Available online: https://www.energy.gov/eere/fuelcells/fuel-cells
[29] J. Kurtz, S. Sprik, G. Saur, and S. Onorato, "On-road fuel cell electric vehicles evaluation: overview," Nat. Renew. Energy Lab. (NREL), Golden, CO, United States, Tech. Rep., no. NREL/TP-5400-73009, Mar. 2019.
[30] Europe fuel cells and hydrogen joint undertaking (FCH 2 JU), "State-of-the-art and future targets (KPIS)," Available online: https://www.fch.europa.eu/soa-and-targets
[31] J. Kurtz, S. Sprik, G. Saur, and S. Onorato, "Fuel cell electric vehicle durability and fuel cell performance," Nat. Renew. Energy Lab. (NREL), Golden, CO, United States, Tech. Rep., no. NREL/TP-5400-73011, Mar. 2019.
[32] Y. Pu, Q. Li, X. Zou, R. Li, L. Li, W. Chen, et al., "Optimal sizing for an integrated energy system considering degradation and seasonal hydrogen storage," *Appl. Energy*, vol. 302, p. 117542, Nov. 2021.
[33] Y. Qiu, Q. Li, L. Huang, C. Sun, T. Wang, and W. Chen, "Adaptive uncertainty sets-based two-stage robust optimization for economic dispatch of microgrid with demand response," *IET Renewable Power Gener.*, vol. 14, Oct. 2020.
[34] P. Ren, P. Pei, Y. Li, Z. Wu, D. Chen, and S. Huang, "Degradation mechanisms of proton exchange membrane fuel cell under typical automotive operating conditions," *Prog. Energy Combust. Sci.,* vol. 80, p. 100859, Sep. 2020.
[35] E. Laffly, M. Pera, and D. Hissel, "Polymer electrolyte membrane fuel cell modelling and parameters estimation for ageing consideration," in *2007 IEEE International Symposium on Industrial Electronics (ISIE)*, Vigo, Spain, pp. 180-185, Jun. 2007.
[36] A. Saadi, M. Becherif, D. Hissel, and H. S. Ramadan, "Dynamic modeling and experimental analysis of PEMFCs: A comparative study," *Int. J. Hydrogen Energy,* vol. 42, no. 2, pp. 1544-1557, Jan. 2017.
[37] J. Li, Z. Hu, L. Xu, M. Ouyang, C. Fang, J. Hu*, et al.*, "Fuel cell system degradation analysis of a Chinese plug-in hybrid fuel cell city bus," *Int. J. Hydrogen Energy,* vol. 41, no. 34, pp. 15295-15310, Sep. 2016.
[38] T. Chu, M. Xie, Y. Yu, B. Wang, D. Yang, B. Li, *et al*., "Experimental study of the influence of dynamic load cycle and operating parameters on the durability of PEMFC," *Energy*, vol. 239, p. 122356, Jan. 2022.
[39] J. Wu, X. Z. Yuan, J. J. Martin, H. Wang, J. Zhang, J. Shen*, et al.*, "A review of PEM fuel cell durability: Degradation mechanisms and mitigation strategies," *J. Power Sources,* vol. 184, no. 1, pp. 104-119, Sep. 2008.
[40] J. Zhao, Z. Tu, and S. H. Chan, "Carbon corrosion mechanism and mitigation strategies in a proton exchange membrane fuel cell (PEMFC): A review," *J. Power Sources*, vol. 488, p. 229434, Mar. 2021.
[41] G. A. Futter, A. Latz, and T. Jahnke, "Physical modeling of chemical membrane degradation in polymer electrolyte membrane fuel cells: Influence of pressure, relative humidity and cell voltage," *J. Power Sources,* vol. 410-411, pp. 78-90, Jan. 2019.
[42] R. L. Borup, A. Kusoglu, K. C. Neyerlin, R. Mukundan, R. K. Ahluwalia, D. A. Cullen, et al., "Recent developments in catalyst-related PEM fuel cell durability," *Curr. Opin. Electrochem*., vol. 21, pp. 192-200, Jun. 2020.
[43] M. Mayur, M. Gerard, P. Schott, and W. Bessler, "Lifetime prediction of a polymer electrolyte membrane fuel cell under automotive load cycling using a physically-based catalyst degradation model," *Energies,* vol. 11, p. 2054, Aug. 2018.
[44] Z. Hu, L. Xu, J. Li, M. Ouyang, Z. Song, and H. Huang, "A reconstructed fuel cell life-prediction model for a fuel cell hybrid city bus," *Energy Convers. Manage.,* vol. 156, pp. 723-732, Jan. 2018.
[45] E. Breaz, F. Gao, A. Miraoui, and R. Tirnovan, "A short review of aging mechanism modeling of proton exchange membrane fuel cell in transportation applications," *in Proc. IECON 40th Annu. Conf. IEEE Ind. Electron. Soc.*, Dallas, TX, USA, pp. 3941-3947, Oct. 2014.
[46] T. Sutharssan, D. Montalvao, Y. K. Chen, W.-C. Wang, C. Pisac, and H. Elemara, "A review on prognostics and health monitoring of proton exchange membrane fuel cell," *Renew. Sustain. Energy Rev.*, vol. 75, pp. 440-450, Aug. 2017.
[47] J. Zhao and X. Li, "A review of polymer electrolyte membrane fuel cell durability for vehicular applications: Degradation modes and experimental techniques," *Energy Convers. Manage.*, vol. 199, p. 112022, Nov. 2019.
[48] International Standard, "Condition monitoring and diagnostics of machines - prognostics - part1: general guidelines," 3rd ed., International Organization for Standardization (ISO), no. ISO13381-1, 2015.
[49] M. Jouin, M. Bressel, S. Morando, R. Gouriveau, D. Hissel, M.-C. Péra, *et al.*, "Estimating the end-of-life of PEM fuel cells: guidelines and metrics," *Appl. Energy*., vol. 177, pp. 87-97, Sep. 2016.
[50] M. Yue, S. Jemei, N. Zerhouni, and R. Gouriveau, "Proton exchange membrane fuel cell system prognostics and decision-making: Current status and perspectives," *Renew. Energy,* vol. 179, Aug. 2021.
[51] R. Gouriveau, M. Hilairet, D. Hissel, S. Jemei, M. Jouin, E. Lechartier, *et al.*, "IEEE PHM 2014 data challenge: outline, experiments, scoring of results, winners, " FCLAB, Belfort, France, Tech. Rep., Jun. 2014.
[52] K. Javed, R. Gouriveau, N. Zerhouni, D. Hissel, "Data-driven prognostics of proton exchange membrane fuel cell stack with constraint based summation-wavelet extreme learning machine," *in Int. Conf. Fundam. Develop. Fuel Cells*, Toulouse, France, pp. 1-8, Feb. 2015.

[53] K. Javed, R. Gouriveau, N. Zerhouni, and D. Hissel, "Improving accuracy of long-term prognostics of PEMFC stack to estimate remaining useful life," *in IEEE International Conference on Industrial Technology (ICIT)*, Seville, Spain, pp. 1047-1052, Mar.2015.

[54] J. Liu, Q. Li, Y. Han, G. Zhang, X. Meng, J. Yu, et al., "PEMFC residual life prediction using sparse autoencoder-based deep neural network," *IEEE Trans. Transp. Electrific.*, vol. 5, no. 4, pp. 1279-1293, Dec. 2019.

[55] S. Morando, S. Jemei, R. Gouriveau, N. Zerhouni, and D. Hissel, "Fuel cells prognostics using echo state network," *in Proc. IECON 39th Annu. Conf. IEEE Ind. Electron. Soc.*, Vienna, Austria, pp. 1632-1637, Nov. 2013.

[56] Z. Hua, Z. Zheng, M.-C. Péra, and F. Gao, "Remaining useful life prediction of PEMFC systems based on the multi-input echo state network," *Appl. Energy.*, vol. 265, p. 114791, May. 2020.

[57] J. K. Kimotho, T. Meyer, and W. Sextro, "PEM fuel cell prognostics using particle filter with model parameter adaptation," *in Proc. ICPHM Int. Conf. Prognostics Health Manage.*, Cheney, WA, USA, pp. 1-6, Jun. 2014.

[58] J. Ma, X. Liu, X. Zou, M. Yue, P. Shang, L. Kang*, et al.*, "Degradation prognosis for proton exchange membrane fuel cell based on hybrid transfer learning and intercell differences," *ISA Trans.,* Jun. 2020.

[59] H. Liu, J. Chen, D. Hissel, M. Hou, and Z. Shao, "A multi-scale hybrid degradation index for proton exchange membrane fuel cells," *J. Power Sources,* vol. 437, p. 226916, Oct. 2019.

[60] J. Spendelow, J. Marcinkoski, and D. Papageorgopoulos, "Micro CHP fuel cell system targets," U.S. DOE, Washington DC, United States, Tech. Rep., no. 11016, Mar. 2012. Available online: https://www.hydrogen.energy.gov/program_records.html

[61] M. Jouin, R. Gouriveau, D. Hissel, M.-C. Péra, and N. Zerhouni, "Prognostics of PEM fuel cell in a particle filtering framework," *Int. J. Hydrogen Energ.*, vol. 39, pp. 481-494, Jan. 2014.

[62] J. Chen, D. Zhou, C. Lyu, and C. Lu, "A novel health indicator for PEMFC state of health estimation and remaining useful life prediction," *Int. J. Hydrogen Energ.*, vol. 42, pp. 20230-20238, Aug. 2017.

[63] D. Zhang, P. Baraldi, C. Cadet, N. Y. Steiner, C. Bérenguer, and E. Zio, "An ensemble of models for integrating dependent sources of information for the prognosis of the remaining useful life of proton exchange membrane fuel cells," *Mech. Syst. and Signal Processing*, vol. 124, pp. 479-501, Jun. 2019.

[64] X. Zhang, P. Pisu, "An unscented Kalman filter based approach for the health-monitoring and prognostics of a electrolyte membrane fuel cell polymer," *Annu. Conf. of PHM Soc.*, vol. 4, no. 1, pp. 1-9, Jan. 2012.

[65] M. Bressel, M. Hilairet, D. Hissel, and B. O. Bouamama, "Remaining useful life prediction and uncertainty quantification of proton exchange membrane fuel cell under variable load," *IEEE Trans. Ind. Electron.*, vol. 63, pp. 2569-2577, Apr. 2016.

[66] Z. Li, S. Jemei, R. Gouriveau, D. Hissel, and N. Zerhouni, "Remaining useful life estimation for PEMFC in dynamic operating conditions," in *proc. IEEE Vehicle Power Propuls. Conf. (VPPC),* Hangzhou, China, pp. 1-6, Oct. 2016.

[67] Z. Hua, Z. Zheng, E. Pahon, M.-C. Péra, and F. Gao, "Remaining useful life prediction of PEMFC systems under dynamic operating conditions," *Energy Convers. Manage.*, vol. 231, p. 113825, Mar. 2021.

[68] Z. Hua, Z. Zheng, E. Pahon, M. C. Péra, and F. Gao, "Health indicators for PEMFC systems life prediction under both static and dynamic operating conditions," *in Proc. IECON 46th Annu. Conf. IEEE Ind. Electron. Soc.*, Singapore, pp. 3963-3968, Oct. 2020.

[69] Hydrogen and Fuel Cell Technologies Office of U.S. DOE, " Hydrogen and fuel cell technologies office multi-year research, development, and demonstration plan (MYRD&D), Fuel cells 2016", Washington DC, United States, May. 2017. Available online: https://www.energy.gov/eere/fuelcells/articles/hydrogen-and-fuel-cell-technologies-office-multi-year-research-development

[70] A. Saxena, J. Celaya, B. Saha, S. Saha, and K. Goebel, "Metrics for offline evaluation of prognostic performance," *Int. J. of Prognostics and Health Manage.*, vol. 1, pp. 2153-2648, Jan. 2010.

[71] H. Liu, J. Chen, Q. Ouyang, and H. Su, "A review on prognostics of proton exchange membrane fuel cells," *in proc. IEEE Vehicle Power Propuls. Conf. (VPPC)*, Hangzhou, China, pp. 1-6, Oct. 2016.

[72] H. Liu, J. Chen, D. Hissel, J. Lu, M. Hou, and Z. Shao, "Prognostics methods and degradation indexes of proton exchange membrane fuel cells: A review," *Renew. Sustain. Energy Rev.,* vol. 123, p. 109721, May. 2020.

[73] R. Onanena, L. Oukhellou, D. Candusso, F. Harel, D. Hissel, and P. Aknin, "Fuel cells static and dynamic characterizations as tools for the estimation of their ageing time," *Int. J. Hydrogen Energ.*, vol. 36, no. 2, pp. 1730-1739, Jan. 2011.

[74] W. O. L. Vianna, I. P. d. Medeiros, B. S. Aflalo, L. R. Rodrigues, and J. P. P. Malère, "Proton Exchange Membrane Fuel Cells (PEMFC) impedance estimation using regression analysis," *in Proc. ICPHM Int. Conf. Prognostics Health Manage.*, Cheney, WA, USA, pp. 1-7, Jun. 2014.

[75] K. Taejin, K. Hyunjae, H. Jongmoon, K. Keunsu, Y. Jungtaek, J. Joonha, et al., "A degenerated equivalent circuit model and hybrid prediction for state-of-health (SOH) of PEM fuel cell," *in Proc. ICPHM Int. Conf. Prognostics Health Manage.*, Cheney, WA, USA, pp. 1-7, Jun. 2014.

[76] E. Lechartier, E. Laffly, M.-C. Péra, R. Gouriveau, D. Hissel, and N. Zerhouni, "Proton exchange membrane fuel cell behavioral model suitable for prognostics," *Int. J. Hydrogen Energ.*, vol. 40, no. 26, pp. 8384-8397, Jul. 2015.

[77] C. Robin, M. Gerard, A. A. Franco, and P. Schott, "Multi-scale coupling between two dynamical models for PEMFC aging prediction," *Int. J. Hydrogen Energ.*, vol. 38, no. 11, pp. 4675-4688, Apr. 2013.

[78] P. Polverino and C. Pianese, "Model-based prognostic algorithm for online RUL estimation of PEMFCs," *in 3rd Conf. Control Fault-Tolerant Syst. (SysTol)*, Barcelona, Spain, pp. 599-604, Sep. 2016.
[79] H. Chen, P. Pei, and M. Song, "Lifetime prediction and the economic lifetime of proton exchange membrane fuel cells," *Appl. Energy.*, vol. 142, pp. 154-163, Mar. 2015.
[80] M. Jouin, R. Gouriveau, D. Hissel, M.-C. Péra, and N. Zerhouni, "Degradations analysis and aging modeling for health assessment and prognostics of PEMFC," *Reliab. Eng. Syst. Saf.*, vol. 148, pp. 78-95, Apr. 2016.
[81] X. Zhang, D. Yang, M. Luo, and Z. Dong, "Load profile based empirical model for the lifetime prediction of an automotive PEM fuel cell," *Int. J. Hydrogen Energ.*, vol. 42, no. 16, pp. 11868-11878, Apr. 2017.
[82] M. Ou, R. Zhang, Z. Shao, B. Li, D. Yang, P. Ming*, et al.*, "A novel approach based on semi-empirical model for degradation prediction of fuel cells," *J. Power Sources,* vol. 488, p. 229435, Mar. 2021.
[83] A. Hochstein, H. I. Ahn, Y. T. Leung, and M. Denesuk, "Switching vector autoregressive models with higher-order regime dynamics," *in Proc. ICPHM Int. Conf. Prognostics Health Manage.*, Cheney, WA, USA, pp. 1-7, Jun. 2014.
[84] D. Zhou, A. Al-Durra, K. Zhang, A. Ravey, and F. Gao, "A robust prognostic indicator for renewable energy technologies: A novel error correction grey prediction model," *IEEE Trans. Ind. Electron.,* vol. 66, no. 12, pp. 9312-9325, Dec. 2019.
[85] B. Feng, M. Fu, H. Ma, Y. Xia, and B. Wang, "Kalman filter with recursive covariance estimation-sequentially estimating process noise covariance," *IEEE Trans. Ind. Electron.,* vol. 61, no. 11, pp. 6253-6263, Nov. 2014.
[86] H. Liu, J. Chen, C. Zhu, H. Su, and M. Hou, "Prognostics of proton exchange membrane fuel cells using a model-based method," *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 4757-4762, Jul. 2017.
[87] R. Pan, D. Yang, Y. Wang, and Z. Chen, "Performance degradation prediction of proton exchange membrane fuel cell using a hybrid prognostic approach," *Int. J. Hydrogen Energy*, vol. 45, no. 55, pp. 30994-31008, Nov. 2020.
[88] K. Chen, S. Laghrouche, and A. Djerdir, "Fuel cell health prognosis using unscented Kalman filter: Postal fuel cell electric vehicles case study," *Int. J. Hydrogen Energy*, vol. 44, no. 3, pp. 1930-1939, Jan. 2019.
[89] C. Higel, F. Harel, D. Candusso, S. Faivre, A. Ravey, D. Guilbert, et al., "Part 1: Mobypost vehicles powertrain modeling, simulation and sizing," *in International Conference on Fundamentals and Development of Fuel Cells(FDFC)*, Karlsruhe, Germany, pp. 1-9, Feb. 2013.
[90] M. Bressel, M. Hilairet, D. Hissel, and B. O. Bouamama, "Fuel cells remaining useful life estimation using an extended Kalman filter," *in Proc. IECON 41st Annu. Conf. IEEE Ind. Electron. Soc.*, Yokohama, Japan, pp. 469-474, Nov. 2015.
[91] M. Bressel, M. Hilairet, D. Hissel, and B. O. Bouamama, "Extended Kalman filter for prognostic of proton exchange membrane fuel cell," *Appl. Energy.*, vol. 164, pp. 220-227, Feb. 2016.
[92] M. Bressel, M. Hilairet, D. Hissel, and B. O. Bouamama, "Fuel cell remaining useful life prediction and uncertainty quantification under an automotive profile," *in Proc. IECON 42nd Annu. Conf. IEEE Ind. Electron. Soc.*, Florence, Italy, pp. 5477-5482, Oct. 2016.
[93] H. Cherragui, M. Bressel, M. Hilairet, and S. Giurgea, "Fuel cells remaining useful life real-time estimation using an extended Kalman filter in a hardware in the loop platform," *in proc. IEEE Vehicle Power Propuls. Conf. (VPPC)*, Belfort, France, pp. 1-6, Dec. 2017.
[94] M. S. Jha, M. Bressel, B. Ould-Bouamama, and G. Dauphin-Tanguy, "Particle filter based hybrid prognostics of proton exchange membrane fuel cell in bond graph framework," *Computers & Chemical Engineering*, vol. 95, pp. 216-230, Nov. 2016.
[95] M. Bressel, D. Hissel, B. Bouamama, and M. Hilairet, "Aging tolerant control of proton exchange membrane fuel cell: a model-based approach," *in Conference on Advanced Control & Diagnosis (ACD)*, Bucarest, Romania, pp. 1-7, Nov. 2017.
[96] M. Bressel, M. Hilairet, D. Hissel, and B. Ould Bouamama, "Model-based aging tolerant control with power loss prediction of proton exchange membrane fuel cell," *Int. J. Hydrogen Energ.*, vol. 45, no. 19, pp. 11242-11254, Apr. 2020.
[97] E. Pahon, S. Morando, R. Petrone, M.-C. Péra, D. Hissel, N. Y. Steiner, et al., "Long-term tests duration reduction for PEMFC μ-CHP application," *Int. J. Hydrogen Energ.*, vol. 42, no. 2, pp. 1527-1533, Jan. 2017.
[98] M. Bressel, M. Hilairet, D. Hissel, and B. Ould Bouamama, "Dynamical modeling of proton exchange membrane fuel cell and parameters identification," *in International Conference on Fundamentals and Development of Fuel Cells(FDFC)*, Toulouse, France, pp. 1-7, Feb. 2015.
[99] Y. Ao, S. Laghrouche, D. Depernet, and K. Chen, "Proton exchange membrane fuel cell prognosis based on frequency domain Kalman filter," *IEEE Trans. Transp. Electrific.*, vol. 7, no. 4, pp. 2332-2343, Dec. 2021.
[100] M. Jouin, R. Gouriveau, D. Hissel, M. Péra, and N. Zerhouni, "Prognostics of proton exchange membrane fuel cell stack in a particle filtering framework including characterization disturbances and voltage recovery," *in Proc. ICPHM Int. Conf. Prognostics Health Manage.*, Cheney, WA, USA, pp. 1-7, Jun. 2014.
[101] M. Jouin, R. Gouriveau, D. Hissel, M.-C. Péra, and N. Zerhouni, "Remaining useful life estimates of a PEM fuel cell stack by including characterization-induced disturbances in a particle filter model," *in Conference Internationale Discussion on Hydrogen Energy and Applications (IDHEA)*, Nantes, France, pp. 1-10, May. 2014.
[102] M. Jouin, R. Gouriveau, D. Hissel, M.-C. Péra, and N. Zerhouni, "Joint particle filters prognostics for proton exchange membrane fuel cell power prediction at constant current solicitation," *IEEE Trans. Reliab.*, vol. 65, no. 1, pp. 336-349, Mar. 2016.

[103] M. Jouin, R. Gouriveau, D. Hissel, M.-C. Péra, and N. Zerhouni, "PEMFC aging modeling for prognostics and health assessment," *IFAC-PapersOnLine*, vol. 48, no. 21, pp. 790-795, Jan. 2015.
[104] M. Jouin, R. Gouriveau, D. Hissel, M. Cécile Péra, and N. Zerhouni, "Combined predictions for prognostics and predictive control of transportation PEMFC," *IFAC-PapersOnLine*, vol. 49, no. 28, pp. 244-249, Jan. 2016.
[105] M. Jouin, R. Gouriveau, D. Hissel, M.-C. Péra, and N. Zerhouni, "Prognostics of PEM fuel cells under a combined heat and power profile," *IFAC-PapersOnLine*, vol. 48, no. 3, pp. 26-31, Jan. 2015.
[106] M. Jouin, R. Gouriveau, D. Hissel, M.-C. Péra, and N. Zerhouni, "PHM of proton-exchange membrane fuel cells - a review," *Chemical Engineering Transactions*, vol. 33, pp. 1009-1015, Dec. 2013.
[107] M. Jouin, R. Gouriveau, D. Hissel, M.-C. Péra, and N. Zerhouni, "Prognostics and health management of PEMFC – state of the art and remaining challenges," *Int. J. Hydrogen Energ.*, vol. 38, no. 35, pp. 15307-15317, Nov. 2013.
[108] M. Jouin, R. Gouriveau, D. Hissel, M.-C. Péra, and N. Zerhouni, "Particle filter-based prognostics: Review, discussion and perspectives," *Mechanical Systems and Signal Processing*, vol. 72-73, pp. 2-31, May. 2016.
[109] D. Zhang, C. Cadet, N. Yousfi-Steiner, and C. Bérenguer, "Proton exchange membrane fuel cell remaining useful life prognostics considering degradation recovery phenomena," *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*, vol. 232, no. 4, pp. 415-424, May. 2018.
[110] D. Zhang, C. Cadet, C. Bérenguer, and N. Yousfi-Steiner, "Some improvements of particle filtering based prognosis for PEM fuel cells," *IFAC-PapersOnLine*, vol. 49, no. 28, pp. 162-167, Oct. 2016.
[111] D. Zhou, Y. Wu, F. Gao, E. Breaz, A. Ravey, and A. Miraoui, "Degradation prediction of PEM fuel cell stack based on multiphysical aging model with particle filter approach," *IEEE Trans. Ind. Appl.*, vol. 53, no. 4, pp. 4041-4052, Jul./Aug. 2017.
[112] P. Wang, H. Liu, J. Chen, X. Qin, W. Lehnert, Z. Shao, et al., "A novel degradation model of proton exchange membrane fuel cells for state of health estimation and prognostics," *Int. J. Hydrogen Energy*, vol. 46, no. 61, pp. 31353-31361, Sep. 2021.
[113] M. Ibrahim, N. Y. Steiner, S. Jemei, and D. Hissel, "Wavelet-based approach for online fuel cell remaining useful lifetime prediction," *IEEE Trans. Ind. Electron.*, vol. 63, no. 8, pp. 5057-5068, Aug. 2016.
[114] D. Zhou, F. Gao, E. Breaz, A. Ravey, and A. Miraoui, "Degradation prediction of PEM fuel cell using a moving window based hybrid prognostic approach," Energy, vol. 138, pp. 1175-1186, Nov. 2017.
[115] D. Zhou, A. Al-Durra, K. Zhang, A. Ravey, and F. Gao, "Online remaining useful lifetime prediction of proton exchange membrane fuel cells using a novel robust methodology," *J. Power Sources*, vol. 399, pp. 314-328, Sep. 2018.
[116] Y. Cheng, N. Zerhouni, and C. Lu, "A hybrid remaining useful life prognostic method for proton exchange membrane fuel cell," *Int. J. Hydrogen Energ.*, vol. 43, no. 27, pp. 12314-12327, Jul. 2018.
[117] L. Zhu and J. Chen, "Prognostics of PEM fuel cells based on Gaussian process state space models," *Energy*, vol. 149, no. 4, pp. 63-73, Apr. 2018.
[118] Y. Xie, J. Zou, C. Peng, Y. Zhu, and F. Gao, "A novel PEM fuel cell remaining useful life prediction method based on singular spectrum analysis and deep Gaussian processes," *Int. J. Hydrogen Energy*, vol. 45, no. 55, pp. 30942-30956, Nov. 2020.
[119] H. Deng, W. Hu, D. Cao, W. Chen, Q. Huang, Z. Chen, et al., "Degradation trajectories prognosis for PEM fuel cell systems based on Gaussian process regression," *Energy*, p. 122569, Nov. 2021.
[120] H. Liu, J. Chen, D. Hissel, and H. Su, "Remaining useful life estimation for proton exchange membrane fuel cells using a hybrid method," *Appl. Energy.*, vol. 237, pp. 910-919, Mar. 2019.
[121] C. Yang, Z. Li, B. Liang, Q. Cui, W. Lu, and J. Yang, "A novel fusion strategy for failure prognostic of proton exchange membrane fuel cell stack," *in 36th Chinese Control Conference (CCC)*, Dalian, China, pp. 7070-7076, Jul. 2017.
[122] C. Yang, Z. Li, B. Liang, W. Lu, X. Wang, and H. Liu, "A particle filter and long short term memory fusion algorithm for failure prognostic of proton exchange membrane fuel cells," *in 29th Chinese Control And Decision Conference (CCDC)*, Chongqing, China, pp. 5646-5651, May. 2017.
[123] R. Ma, Z. Li, E. Breaz, C. Liu, H. Bai, P. Briois, et al., "Data-fusion prognostics of proton exchange membrane fuel cell degradation," *IEEE Trans. Ind. Appl*., vol. 55, no. 4, pp. 4321-4331, July-Aug. 2019.
[124] R. Ma, R. Xie, L. Xu, Y. Huangfu, and Y. Li, "A hybrid prognostic method for PEMFC with aging parameter prediction," *IEEE Trans. Transp. Electrific*., vol. 7, no. 4, pp. 2318-2331, Dec. 2021.
[125] J. Zuo, H. Lv, D. Zhou, Q. Xue, L. Jin, W. Zhou*, et al.*, "Deep learning based prognostic framework towards proton exchange membrane fuel cell for automotive application," *Appl. Energy,* vol. 281, p. 115937, Jan. 2021.
[126] M. Yue, Z. Li, R. Roche, S. Jemei, and N. Zerhouni, "Degradation identification and prognostics of proton exchange membrane fuel cell under dynamic load," *Control Eng. Pract.,* vol. 118, p. 104959, Jan. 2022.
[127] C. Wang, Z. Li, R. Outbib, M. Dou, and D. Zhao, "Symbolic deep learning based prognostics for dynamic operating proton exchange membrane fuel cells," *Appl. Energy*, vol. 305, p. 117918, Jan. 2022.
[128] R. E. Silva, R. Gouriveau, S. Jemeï, D. Hissel, L. Boulon, K. Agbossou, et al., "Proton exchange membrane fuel cell degradation prediction based on adaptive neuro-fuzzy inference systems," *Int. J. Hydrogen Energ*., vol. 39, no. 21, pp. 11128-11144, Jul. 2014.

[129] L. Mao and L. M. Jackson, "Comparative study on prediction of fuel cell performance using machine learning approaches," *in Proceedings of the International MultiConference of Engineers and Computer Scientists (IMECM)*, Hong Kong, China, pp. 1-6, Mar. 2016.
[130] H. Liu, J. Chen, D. Hissel, and H. Su, "Short-term prognostics of PEM fuel cells: a comparative and improvement study," *IEEE Trans. Ind. Electron*., vol. 66, no. 8, pp. 6077-6086, Aug. 2019.
[131] Y. Xie, J. Zou, Z. Li, F. Gao, and C. Peng, "A novel deep belief network and extreme learning machine based performance degradation prediction method for proton exchange membrane fuel cell," *IEEE Access*, vol. 8, pp. 176661-176675, Oct. 2020.
[132] K. Javed, R. Gouriveau, N. Zerhouni, and D. Hissel, "Prognostics of proton exchange membrane fuel cells stack using an ensemble of constraints based connectionist networks," *J. Power Sources*, vol. 324, pp. 745-757, Aug. 2016.
[133] K. Javed, R. Gouriveau, N. Zerhouni, and D. Hissel, "PEM fuel cell prognostics under variable load: a data-driven ensemble with new incremental learning," *in Proc. Int. Conf. Control, Decis. Inf. Technol. (CoDIT)*, Saint Julians, Malta, pp. 252-257, Apr. 2016.
[134] H. Liu, J. Chen, M. Hou, Z. Shao, and H. Su, "Data-based short-term prognostics for proton exchange membrane fuel cells," *Int. J. Hydrogen Energ*., vol. 42, pp. 20791-20808, Aug. 2017.
[135] X. Zhang, Z. Yu, and W. Chen, "Life prediction based on D-S ELM for PEMFC," *Energies*, vol. 12, no. 19, Sep. 2019.
[136] K. Chen, S. Laghrouche, and A. Djerdir, "Degradation model of proton exchange membrane fuel cell based on a novel hybrid method," *Appl. Energy*., vol. 252, p. 113439, Oct. 2019.
[137] K. Chen, S. Laghrouche, and A. Djerdir, "Prognosis of fuel cell degradation under different applications using wavelet analysis and nonlinear autoregressive exogenous neural network," *Renew. Energy,* vol. 179, pp. 802-814, Dec. 2021.
[138] Z. Hua, Z. Zheng, E. Pahon, M.C. Péra, and F. Gao, "Multi-timescale lifespan prediction for PEMFC systems under dynamic operating conditions," *IEEE Trans. Transp. Electrific.*, PP. 1-1, Aug. 2021.
[139] K. Chen, S. Laghrouche, and A. Djerdir, "Degradation prediction of proton exchange membrane fuel cell based on grey neural network model and particle swarm optimization," *Energy Convers. Manage*., vol. 195, pp. 810-818, May. 2019.
[140] K. Chen, S. Laghrouche, and A. Djerdir, "Aging prognosis model of proton exchange membrane fuel cell in different operating conditions," *Int. J. Hydrogen Energ*., vol. 45, no. 20, pp. 11761-11772, Apr.2020.
[141] B. Long, K. Wu, P. Li, and M. Li, "A novel remaining useful life prediction method for hydrogen fuel cells based on the gated recurrent unit neural network," *Appl. Sci.*, vol. 12, no. 1, 2022.
[142] Y. Wu, E. Breaz, F. Gao, and A. Miraoui, "A modified relevance vectormachine for PEM fuel-cell stack aging prediction," *IEEE Trans. Ind. Appl*., vol. 52, no.3, pp. 2573-2581, May. 2016.
[143] Y. Wu, E. Breaz, F. Gao, D. Paire, and A. Miraoui, "Nonlinear performance degradation prediction of proton exchange membrane fuel cells using relevance vector machine," *IEEE Trans. Energy Convers*, vol. 31, no. 4, pp. 1570-1582, Dec. 2016.
[144] R. Ma, T. Yang, E. Breaz, Z. Li, P. Briois, and F. Gao, "Data-driven proton exchange membrane fuel cell degradation predication through deep learning method," *Appl. Energy*., vol. 231, pp. 102-115, Dec. 2018.
[145] R. Ma, E. Breaz, C. Liu, H. Bai, P. Briois, and F. Gao, "Data-driven prognostics for pem fuel cell degradation by long short-term memory network," *in Proc. ITEC IEEE Transp. Electrif. Conf. Expo.*, Long Beach, CA, USA, pp. 102-107, Jun. 2018.
[146] J. Liu, Q. Li, W. Chen, Y. Yan, Y. Qiu, and T. Cao, "Remaining useful life prediction of PEMFC based on long short-term memory recurrent neural networks," *Int. J. Hydrogen Energ.*, vol. 44, no. 11, pp. 5470-5480, Feb. 2019.
[147] F.-K. Wang, X.-B. Cheng, and K.-C. Hsiao, "Stacked long short-term memory model for proton exchange membrane fuel cell systems degradation," *J. Power Sources,* vol. 448, p. 227591, Feb. 2020.
[148] F.-K. Wang, T. Mamo, and X.-B. Cheng, "Bi-directional long short-term memory recurrent neural network with attention for stack voltage degradation from proton exchange membrane fuel cells," *J. Power Sources*, vol. 461, p. 228170, Jun. 2020.
[149] R. Xie, R. Ma, S. Pu, L. Xu, D. Zhao, and Y. Huangfu, "Prognostic for fuel cell based on particle filter and recurrent neural network fusion structure," *Energy and AI*, vol. 2, p. 100017, Nov. 2020.
[150] C. Wang, Z. Li, R. Outbib, M. Dou, and D. Zhao, "A novel long short-term memory networks-based data-driven prognostic strategy for proton exchange membrane fuel cells," *Int. J. Hydrogen Energy*, Jan. 2022.
[151] S. Morando, S. Jemei, D. Hissel, R. Gouriveau, and N. Zerhouni, "Predicting the remaining useful lifetime of a proton exchange membrane fuel cell using an echo state network," *in International Discussion on Hydrogen Energy and Applications (IDHEA)*, Nantes, France, pp. 1-9, May. 2014.
[152] S. Morando, S. Jemei, R. Gouriveau, N. Zerhouni, and D. Hissel, "Fuel cells remaining useful lifetime forecasting using echo state network," *in proc. IEEE Vehicle Power Propuls. Conf. (VPPC)*, Coimbra, Portugal, pp. 1-6, Oct. 2014.
[153] S. Morando, S. Jemei, D. Hissel, R. Gouriveau, and N. Zerhouni, "Proton exchange membrane fuel cell ageing forecasting algorithm based on echo state network," *Int. J. Hydrogen Energ*., vol. 42, no.2, pp. 1472-1480, Jan. 2017.
[154] S. Morando, S. Jemei, D. Hissel, R. Gouriveau, and N. Zerhouni, "ANOVA method applied to proton exchange membrane fuel cell ageing forecasting using an echo state network," *Math. Comput. Simul.*, vol. 131, pp. 283-294, Jan. 2017.

[155] R. Mezzi, S. Morando, N. Y. Steiner, M.-C. Péra, D. Hissel, and L. Larger, "Multi-reservoir echo state network for proton exchange membrane fuel cell remaining useful life prediction," *in Proc. IECON 44th Annu. Conf. IEEE Ind. Electron. Soc.*, Washington, DC, USA, pp. 1872-1877, Oct. 2018.

[156] Z. Li, Z. Zheng, and R. Outbib, "Adaptive prognostic of fuel cells by implementing ensemble echo state networks in time varying model space, " *IEEE Trans. Ind. Electron*., vol. 67, no. 1, pp. 379-389, Jan. 2020.

[157] Z. Hua, Z. Zheng, M.-C. Péra, and F. Gao, "Data-driven prognostics for PEMFC systems by different echo state network prediction structures," *in Proc. ITEC IEEE Transp. Electrif. Conf. Expo.*, Chicago, IL, USA, pp. 495-500, Jun. 2020.

[158] J. Jin, Y. Chen, Z. Wenchao, C. Xie, and F. Wu, "Remaining useful life prediction of PEMFC based on cycle reservoir with jump model," *Int. J. Hydrogen Energy,* vol. 46, no. 80, pp. 40001-40013, Nov. 2021.

[159] Z. Deng, Q. Chen, L. Zhang, K. Zhou, Y. Zong, H. Liu, *et al*., "Degradation prediction of PEMFCs using stacked echo state network based on genetic algorithm optimization," *IEEE Trans. Transp. Electrif.*, pp. 1-1, 2021.

[160] R. Mezzi, N. Yousfi-Steiner, M.-C. Péra, D. Hissel, and L. Larger, "An echo state network for fuel cell lifetime prediction under a dynamic micro-cogeneration load profile," *Appl. Energy*., vol. 283, p. 116297, Feb. 2021.

[161] Z. Hua, Z. Zheng, E. Pahon, M. C. Péra, and F. Gao, "Lifespan prediction for proton exchange membrane fuel cells based on wavelet transform and echo state network," *IEEE Trans. Transp. Electrif.,* pp. 1-1, Oct. 2021.

[162] L. Vichard, F. Harel, A. Ravey, P. Venet, and D. Hissel, "Degradation prediction of PEM fuel cell based on artificial intelligence," *Int. J. Hydrogen Energ.*, vol. 45, no. 29, pp. 14953-14963, May. 2020.

[163] K. He, L. Mao, J. Yu, W. Huang, Q. He, and L. Jackson, "Long-term performance prediction of PEMFC based on LASSO-ESN," *IEEE Trans. Instrum. Meas.,* vol. 70, pp. 1-11, Feb. 2021.

[164] R. Gouriveau and N. Zerhouni, "Connexionist-systems-based long term prediction approaches for prognostics," *IEEE Trans. Reliab.,* vol. 61, no. 4, pp. 909-920, Dec. 2012.